

BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

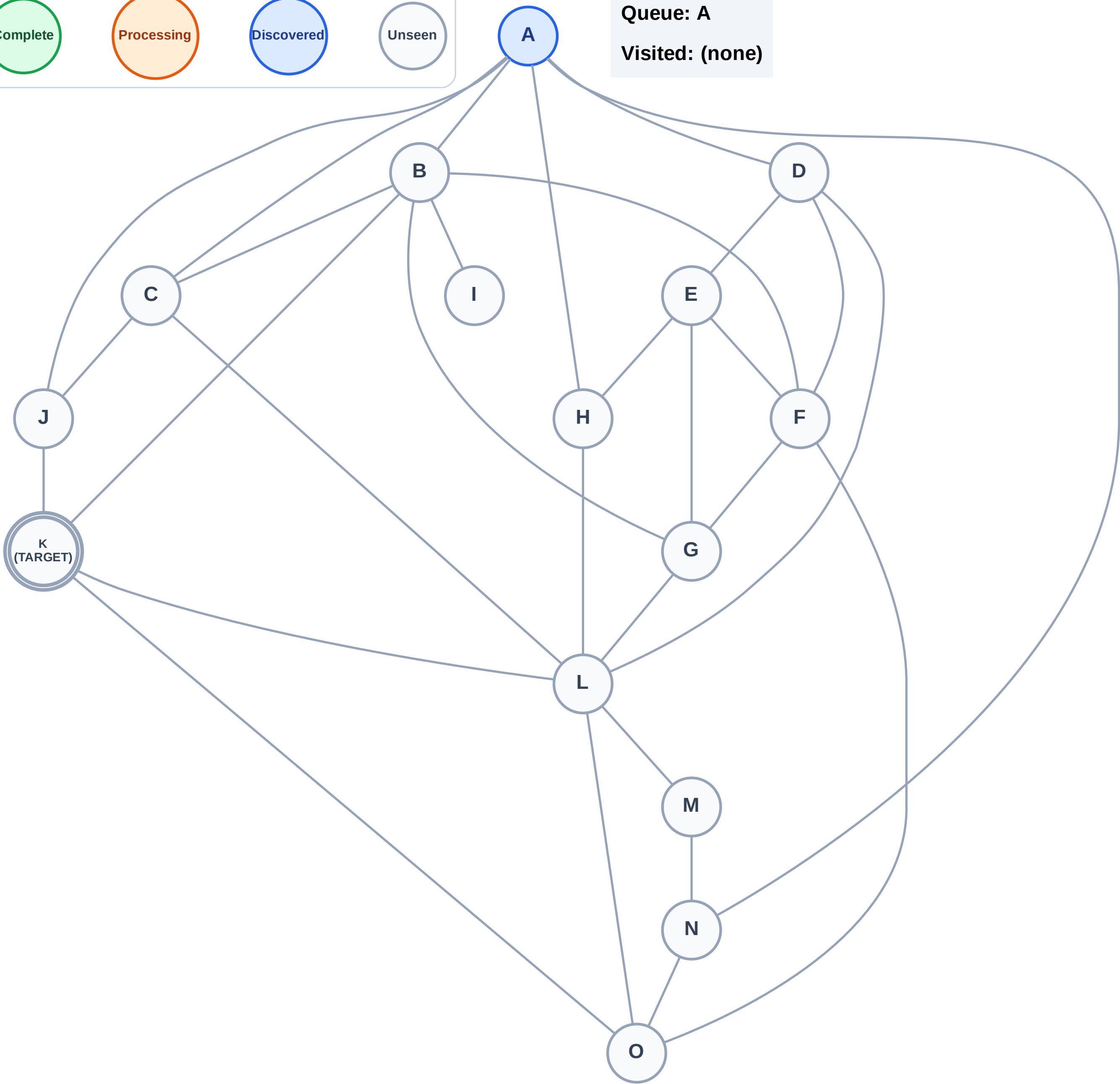
Complete

Processing

Discovered

Unseen

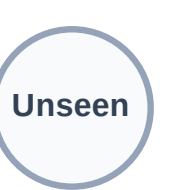
Queue: A
Visited: (none)



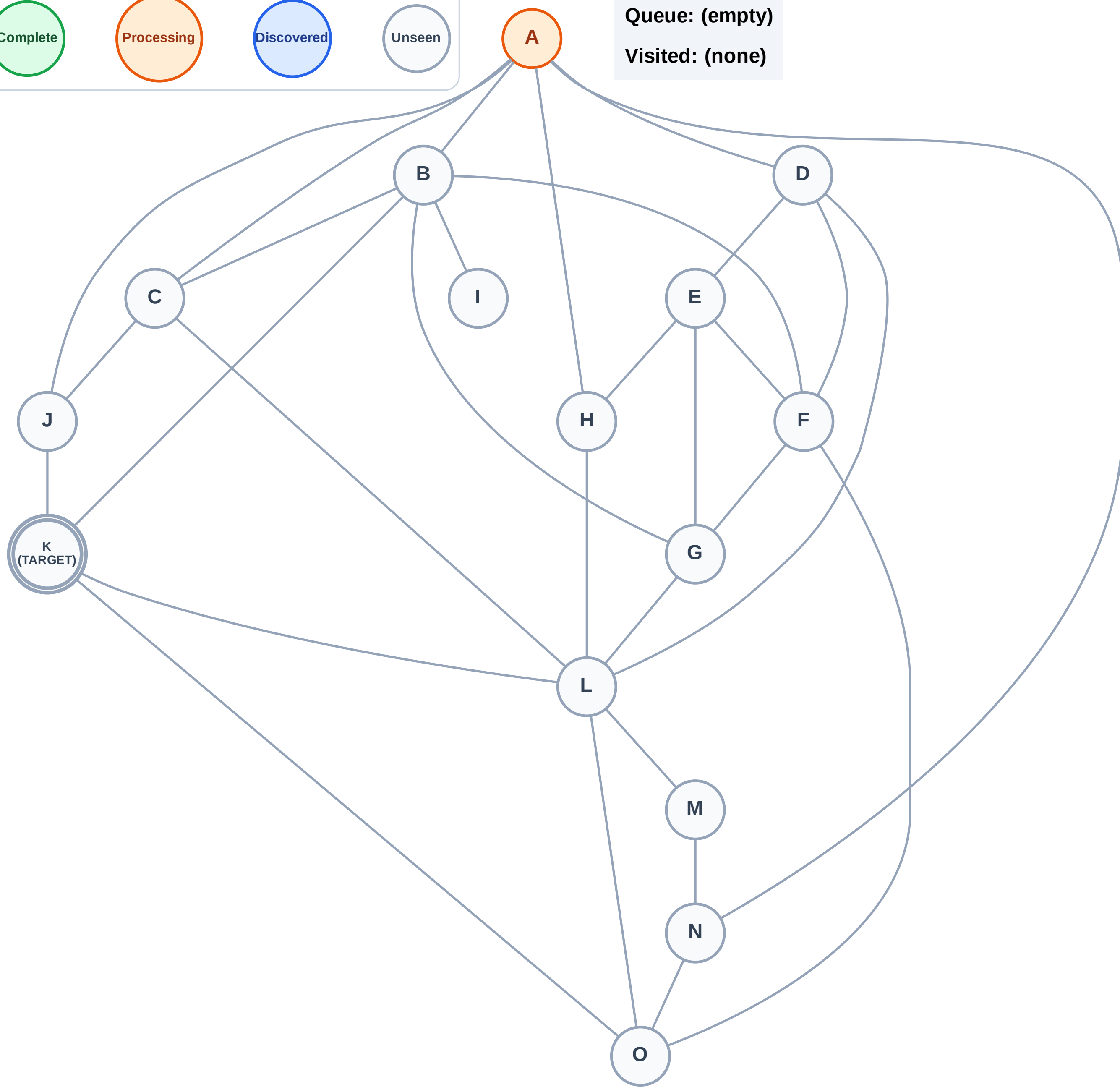
BFS Traversal

Step 2: Process node A

Legend



Queue: (empty)
Visited: (none)



BFS Traversal

Step 3: Discover B, C, D, H, J, N from A

Legend

Complete

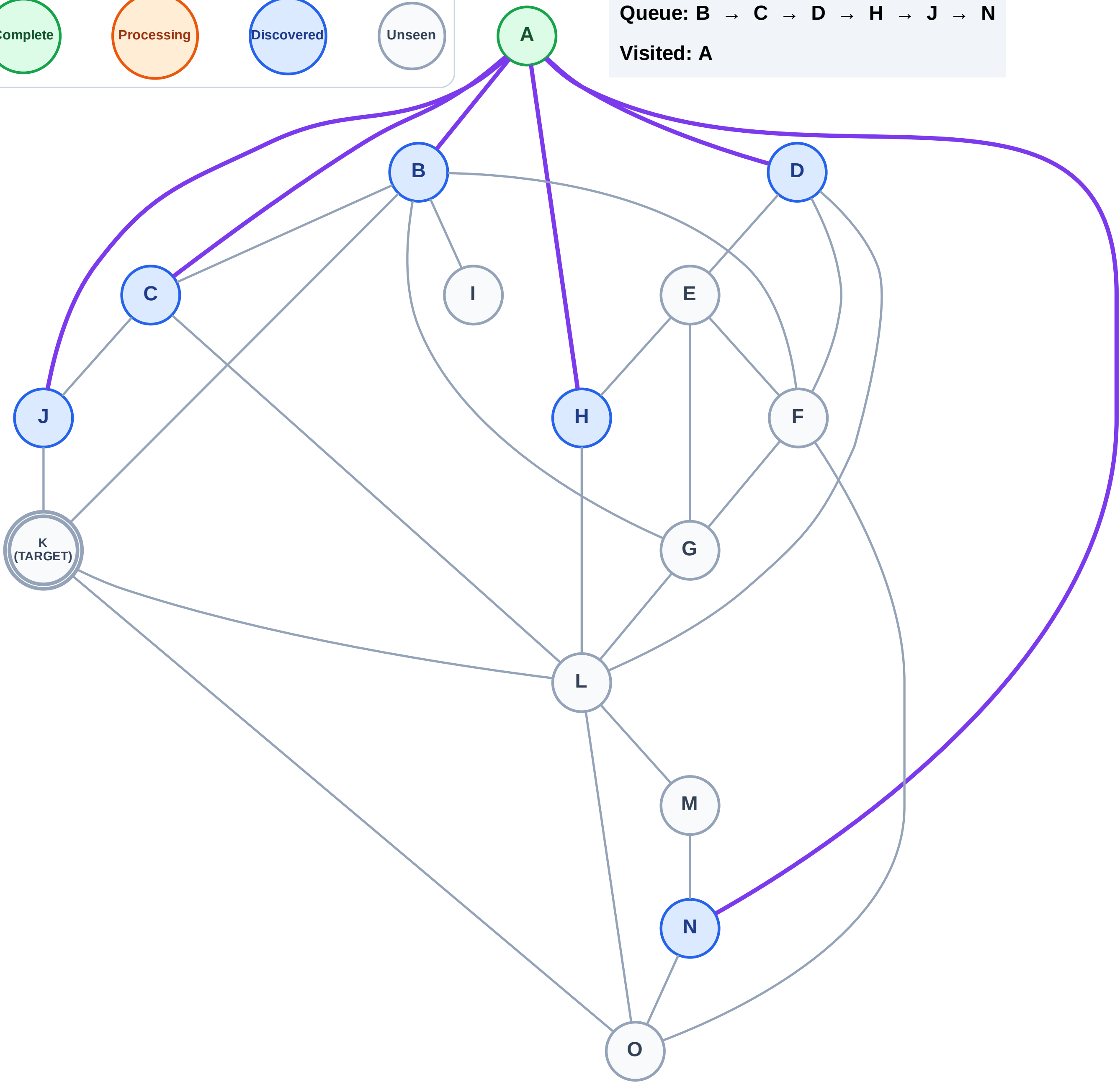
Processing

Discovered

Unseen

Queue: B → C → D → H → J → N

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

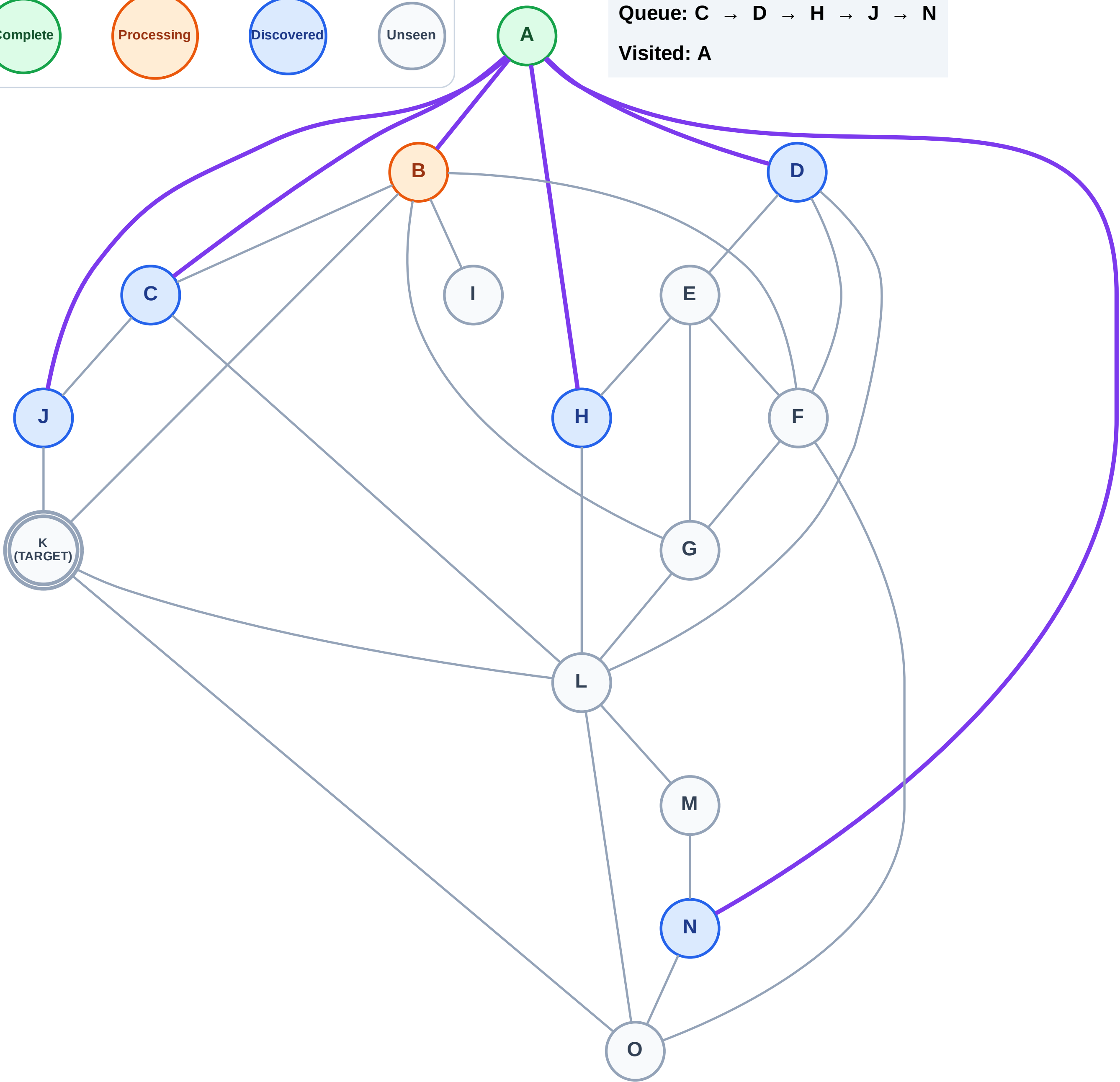
Processing

Discovered

Unseen

Queue: C → D → H → J → N

Visited: A



BFS Traversal

Step 5: Discover F, G, I, K from B

Legend

Complete

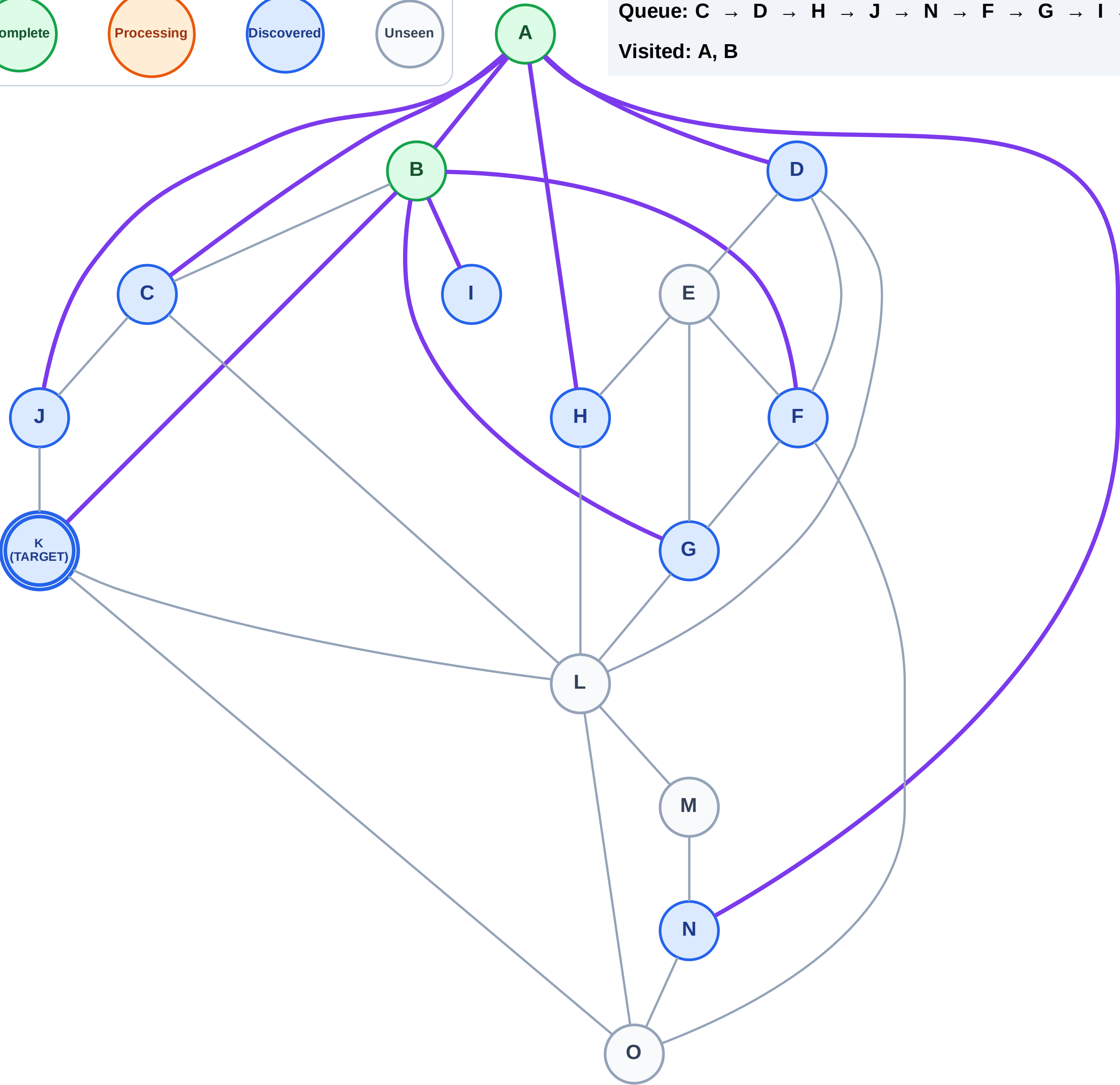
Processing

Discovered

Unseen

Queue: C → D → H → J → N → F → G → I → K

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

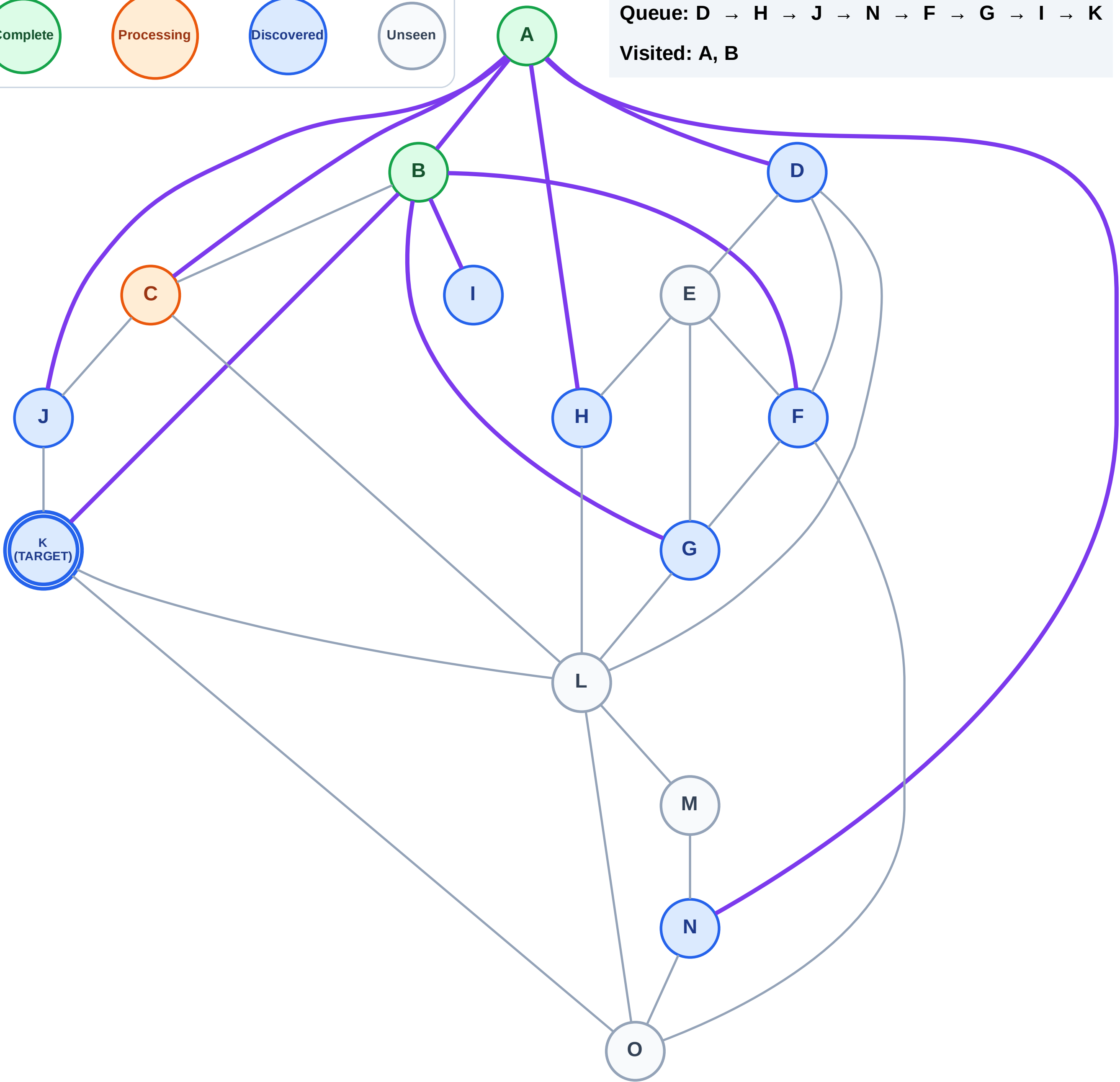
Processing

Discovered

Unseen

Queue: D → H → J → N → F → G → I → K

Visited: A, B



BFS Traversal

Step 7: Discover L from C

Legend

Complete

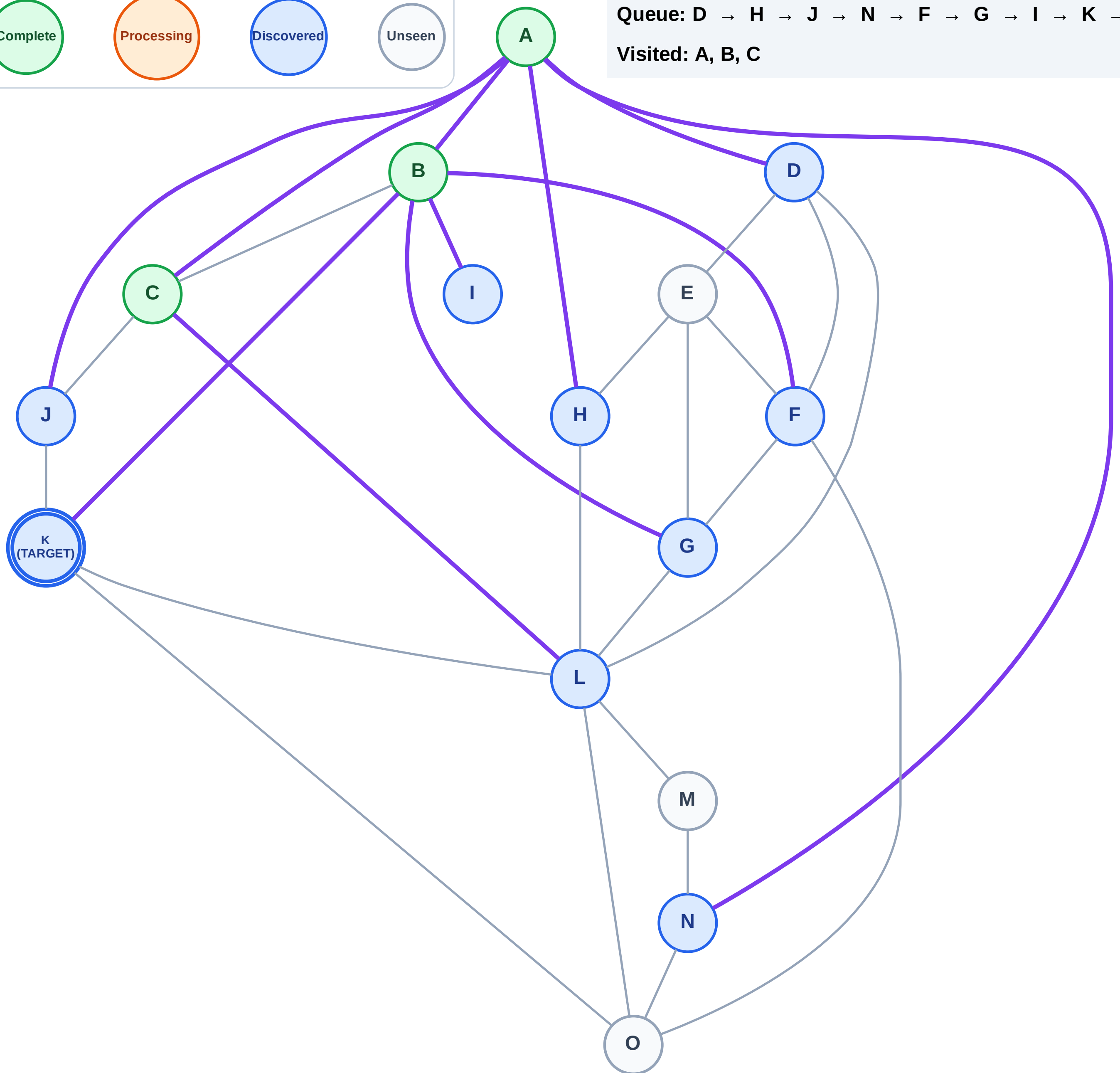
Processing

Discovered

Unseen

Queue: D → H → J → N → F → G → I → K → L

Visited: A, B, C



BFS Traversal

Step 8: Process node D

Legend

Complete

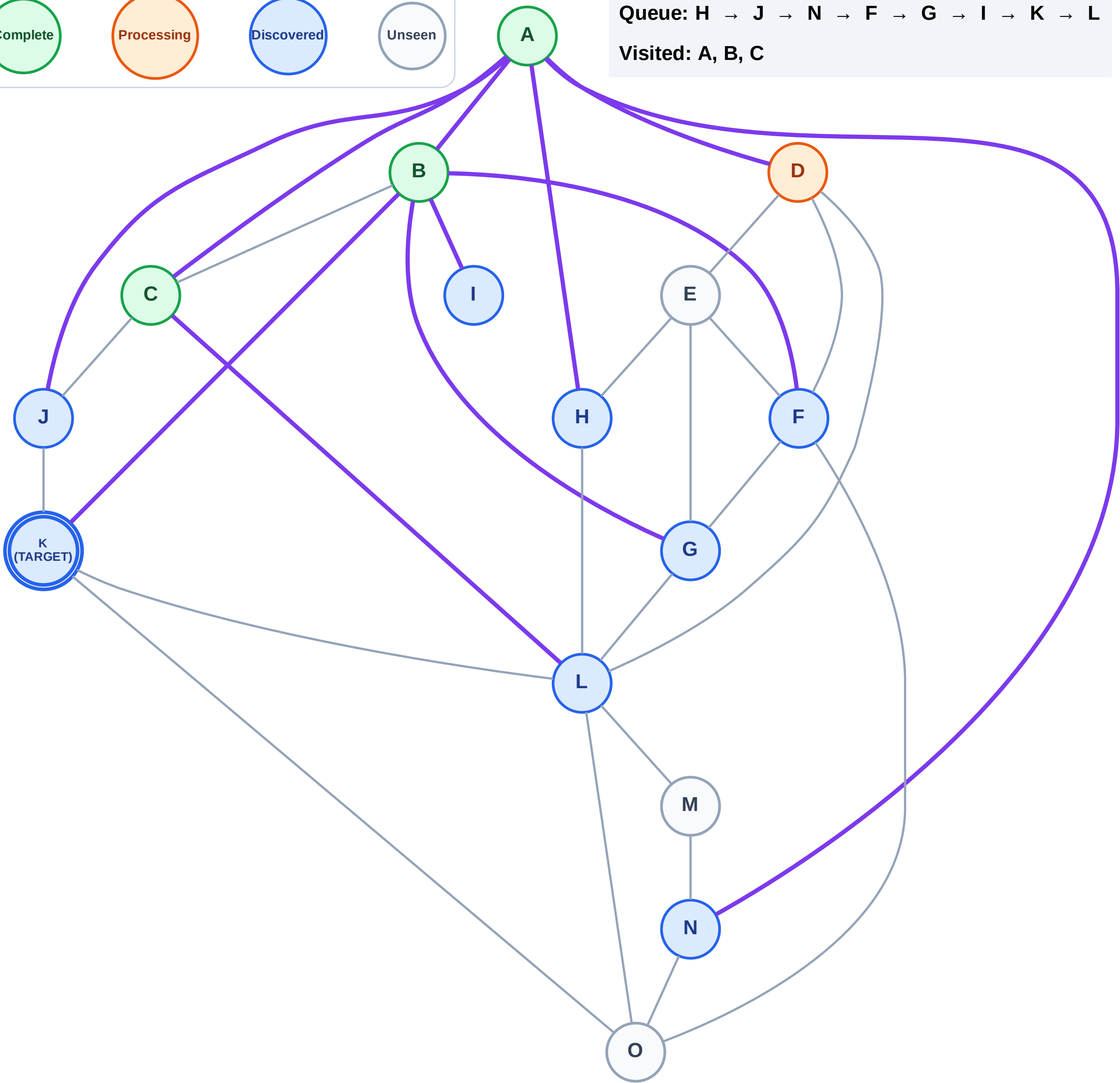
Processing

Discovered

Unseen

Queue: H → J → N → F → G → I → K → L

Visited: A, B, C



BFS Traversal

Step 9: Discover E from D

Legend

Complete

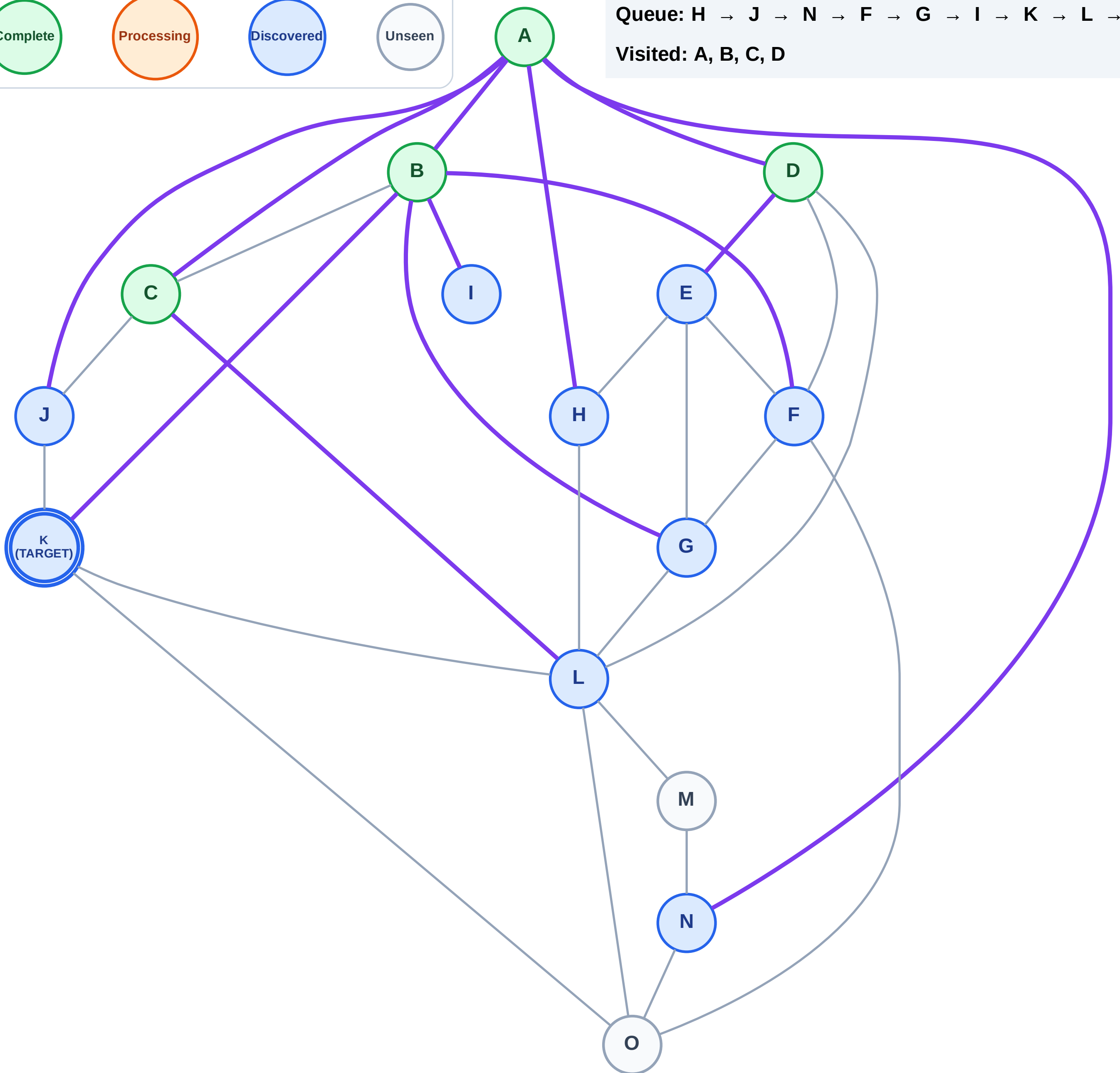
Processing

Discovered

Unseen

Queue: H → J → N → F → G → I → K → L → E

Visited: A, B, C, D



BFS Traversal

Step 10: Process node H

Legend

Complete

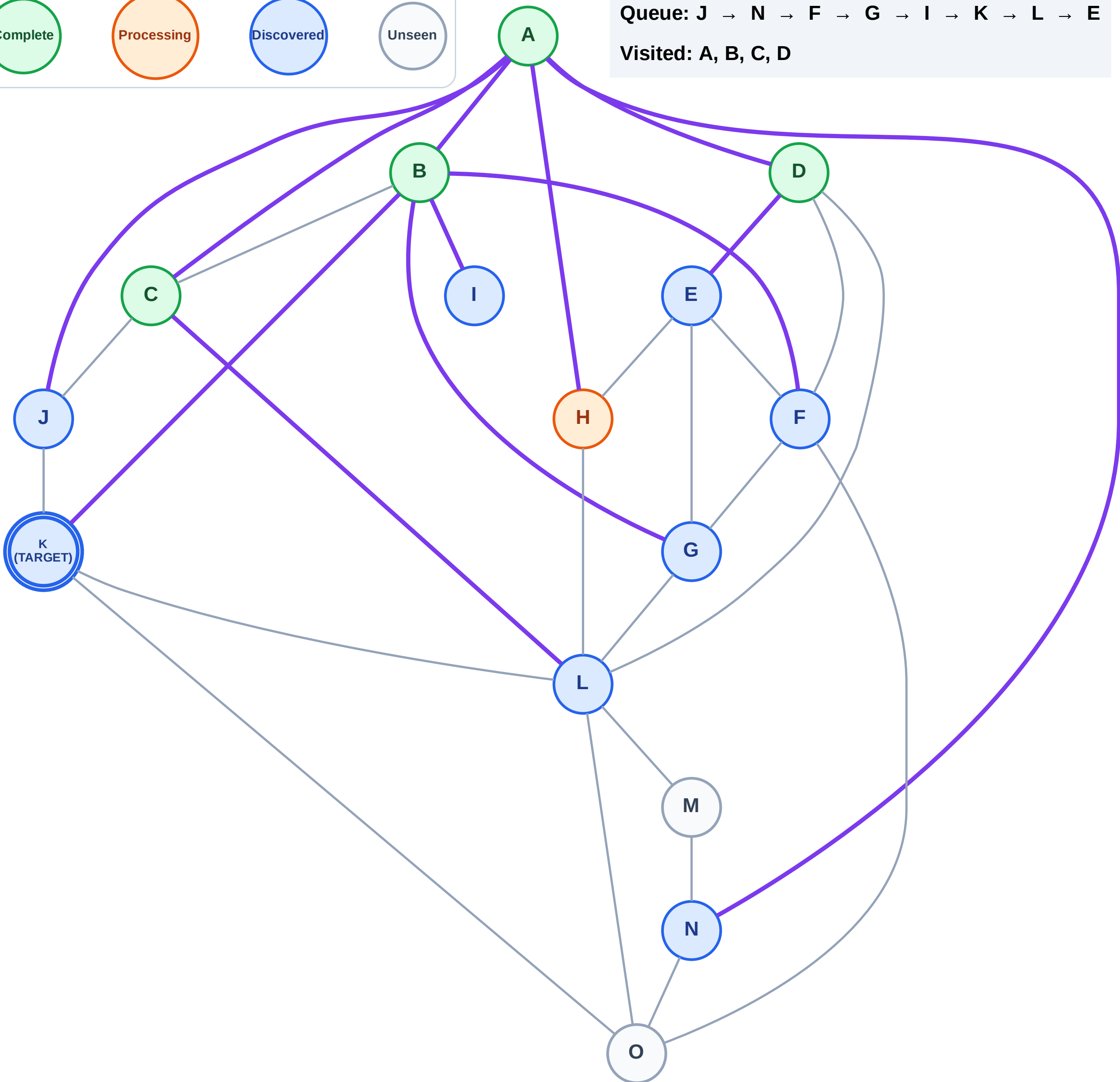
Processing

Discovered

Unseen

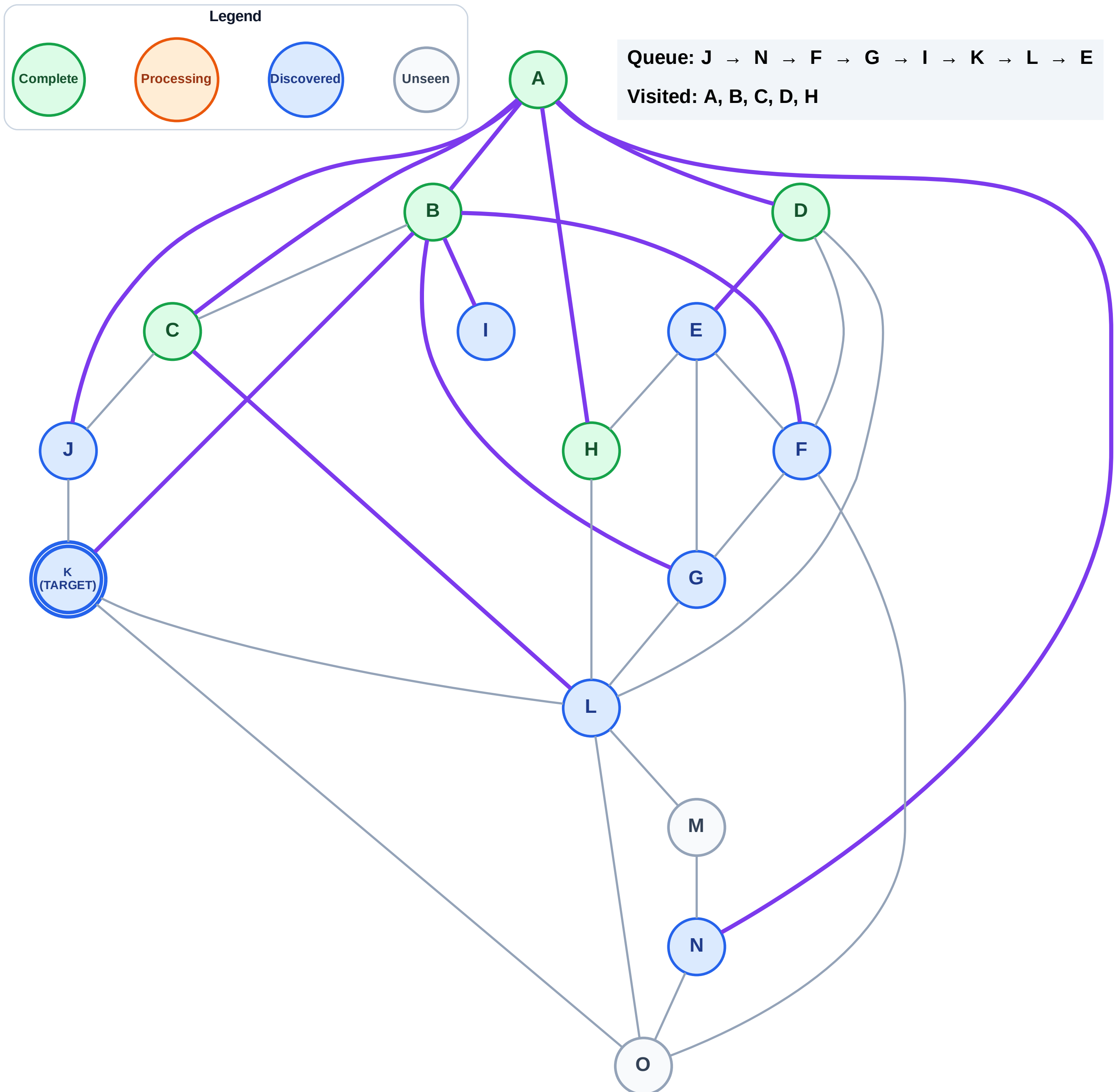
Queue: J → N → F → G → I → K → L → E

Visited: A, B, C, D



Step 11: No new neighbors from H

Step 11: No new neighbors from H



BFS Traversal

Step 12: Process node J

Legend

Complete

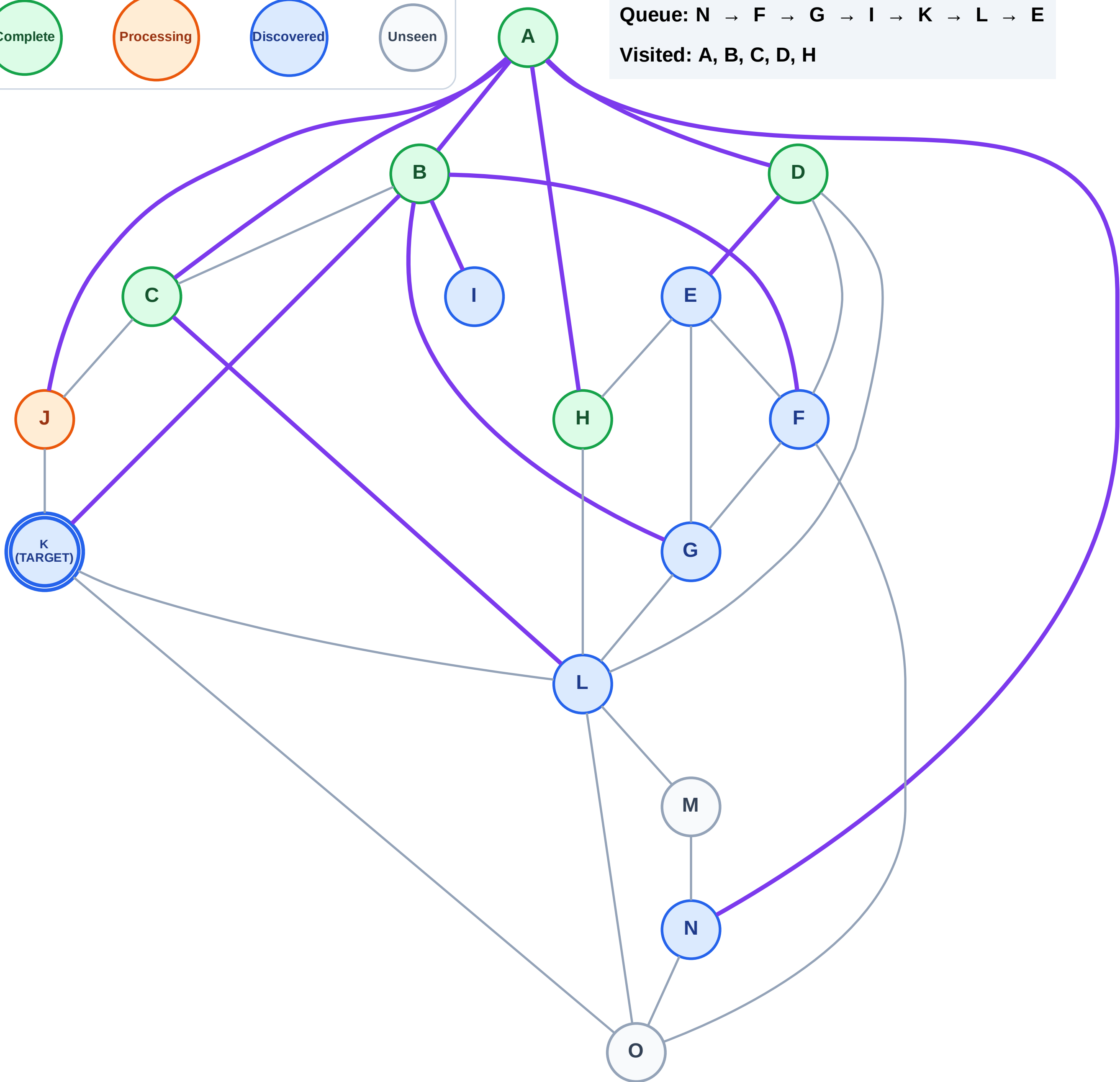
Processing

Discovered

Unseen

Queue: N → F → G → I → K → L → E

Visited: A, B, C, D, H



BFS Traversal

Step 13: No new neighbors from J

Legend

Complete

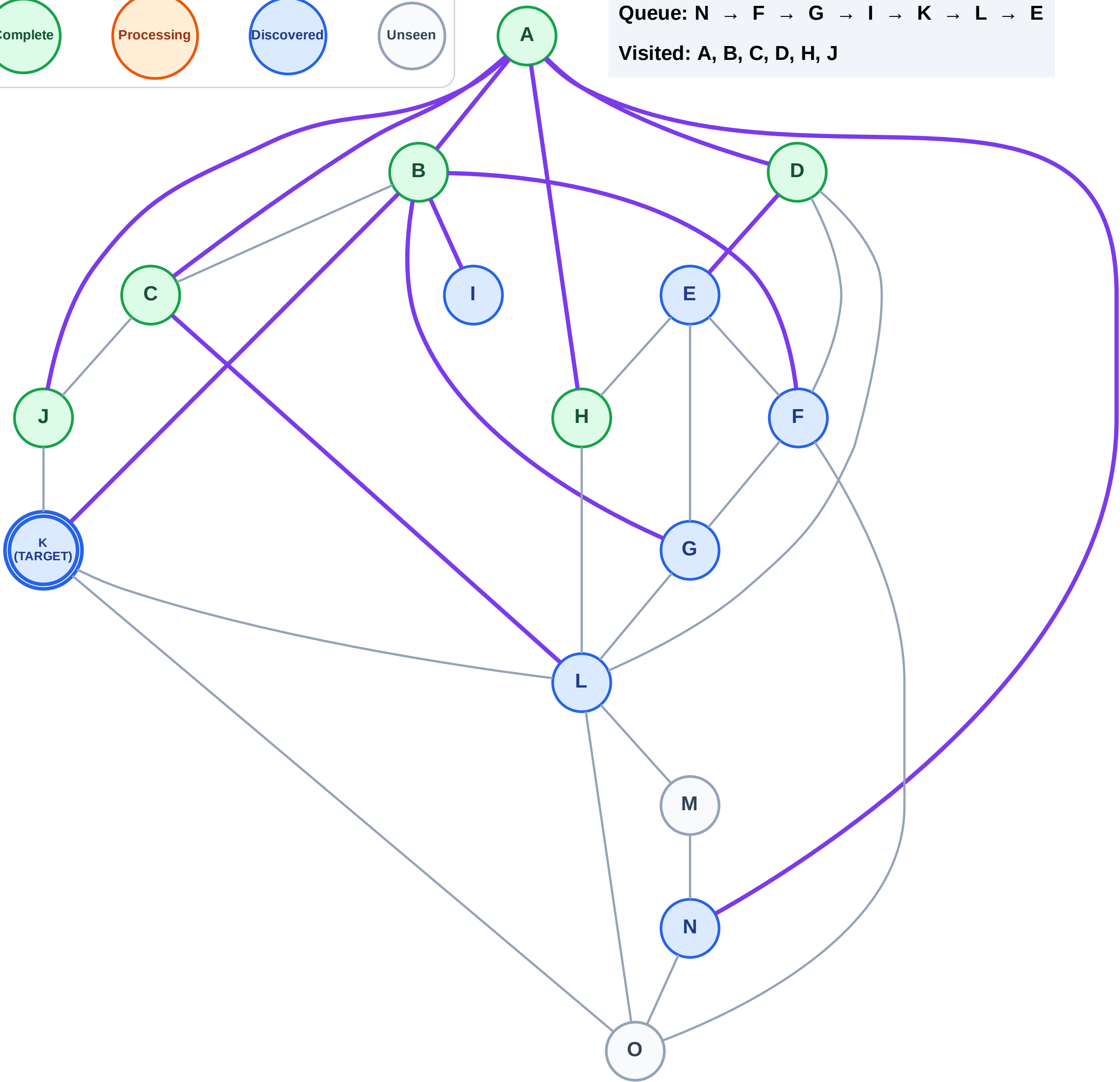
Processing

Discovered

Unseen

Queue: N → F → G → I → K → L → E

Visited: A, B, C, D, H, J



BFS Traversal

Step 14: Process node N

Legend

Complete

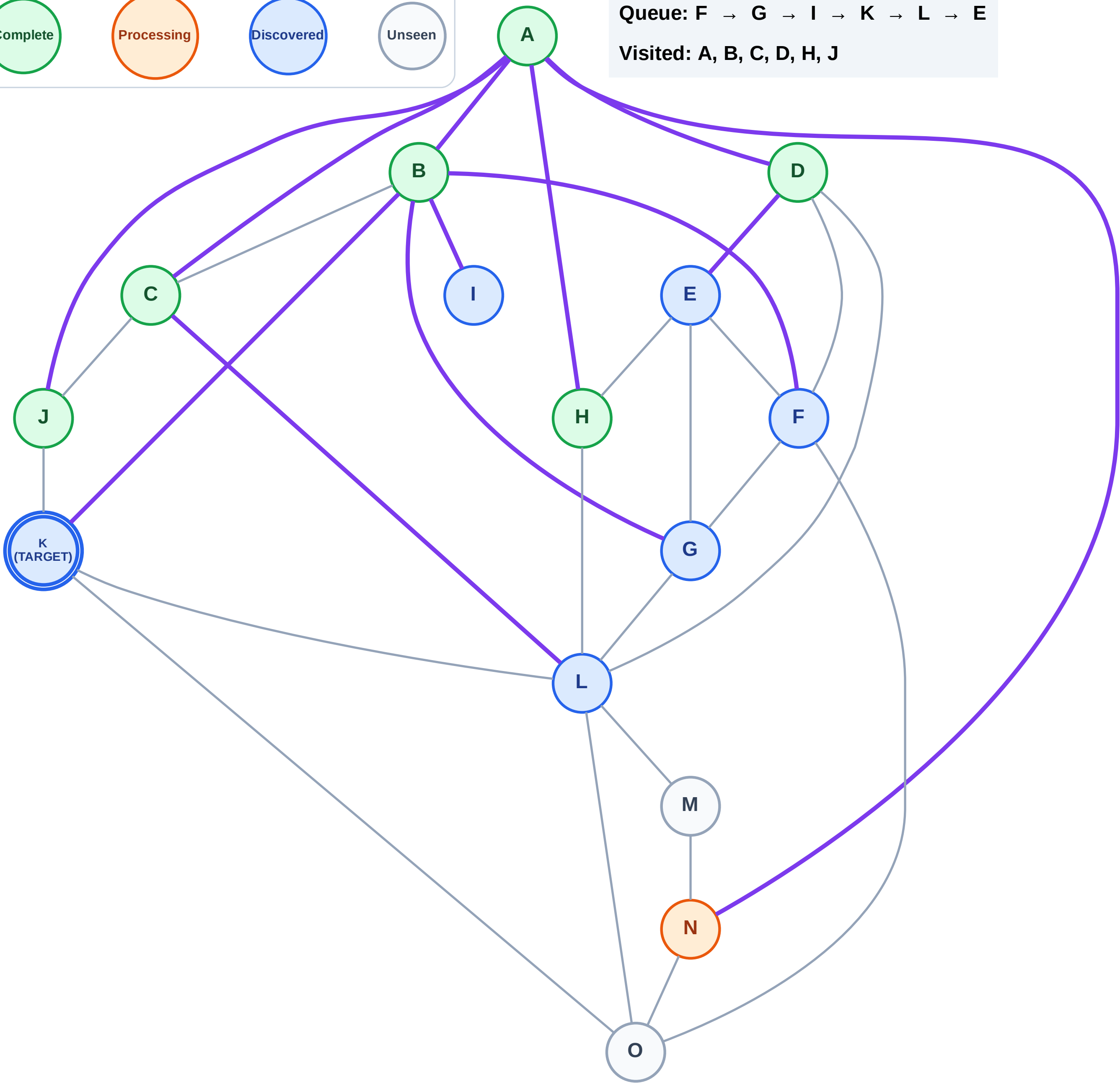
Processing

Discovered

Unseen

Queue: F → G → I → K → L → E

Visited: A, B, C, D, H, J



BFS Traversal

Step 15: Discover M, O from N

Legend

Complete

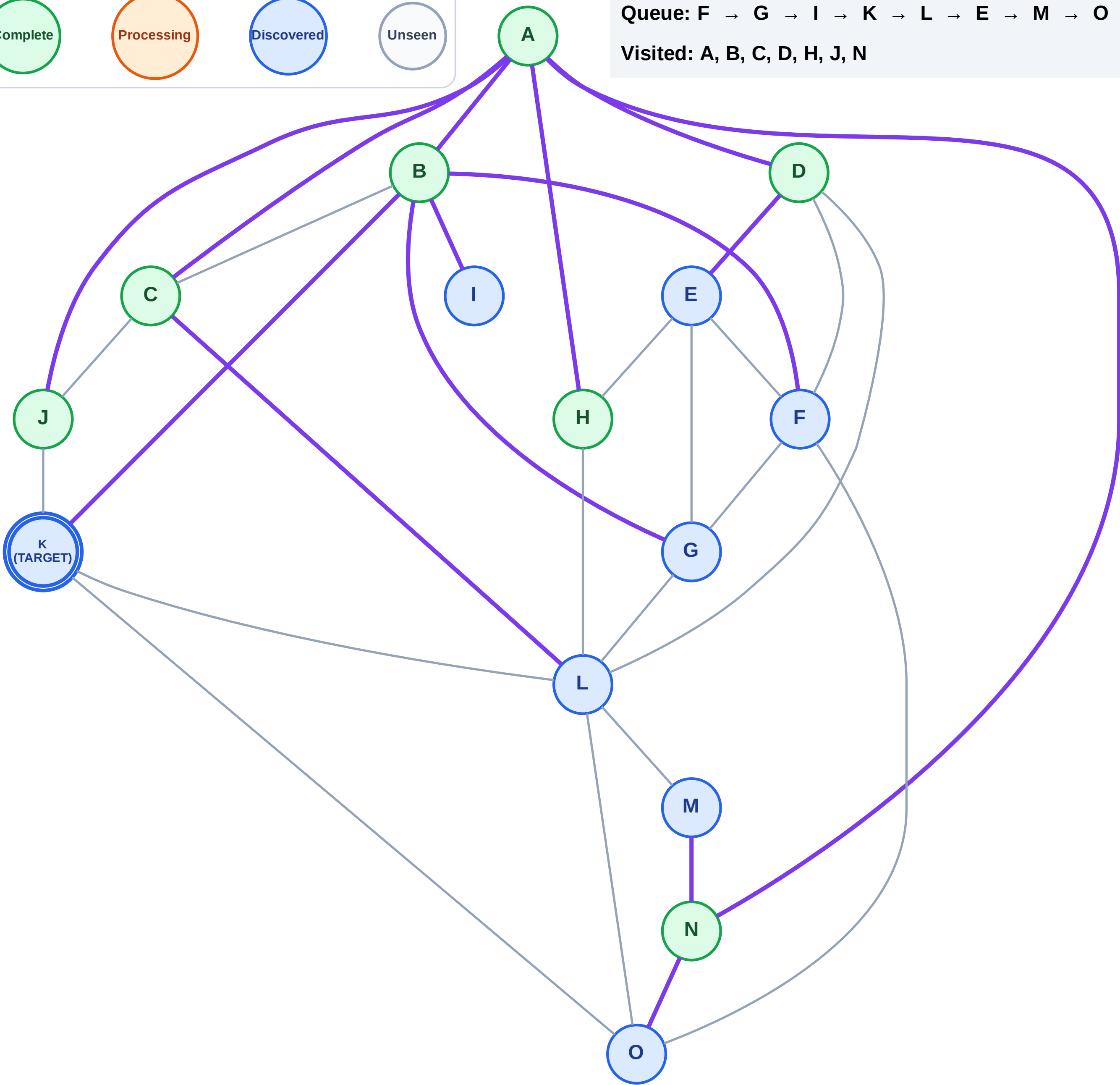
Processing

Discovered

Unseen

Queue: F → G → I → K → L → E → M → O

Visited: A, B, C, D, H, J, N



BFS Traversal

Step 16: Process node F

Legend

Complete

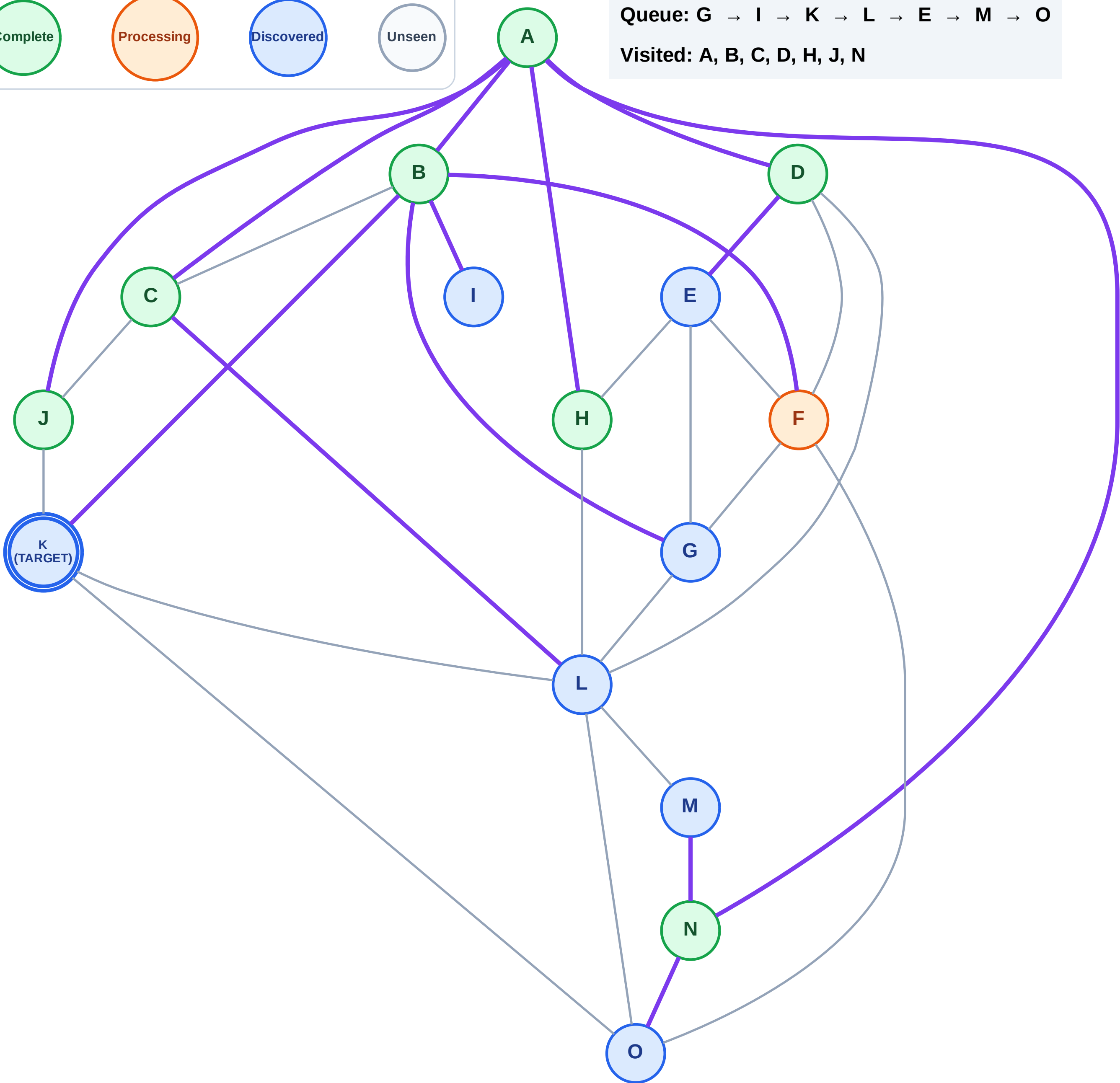
Processing

Discovered

Unseen

Queue: G → I → K → L → E → M → O

Visited: A, B, C, D, H, J, N



BFS Traversal

Step 17: No new neighbors from F

Legend

Complete

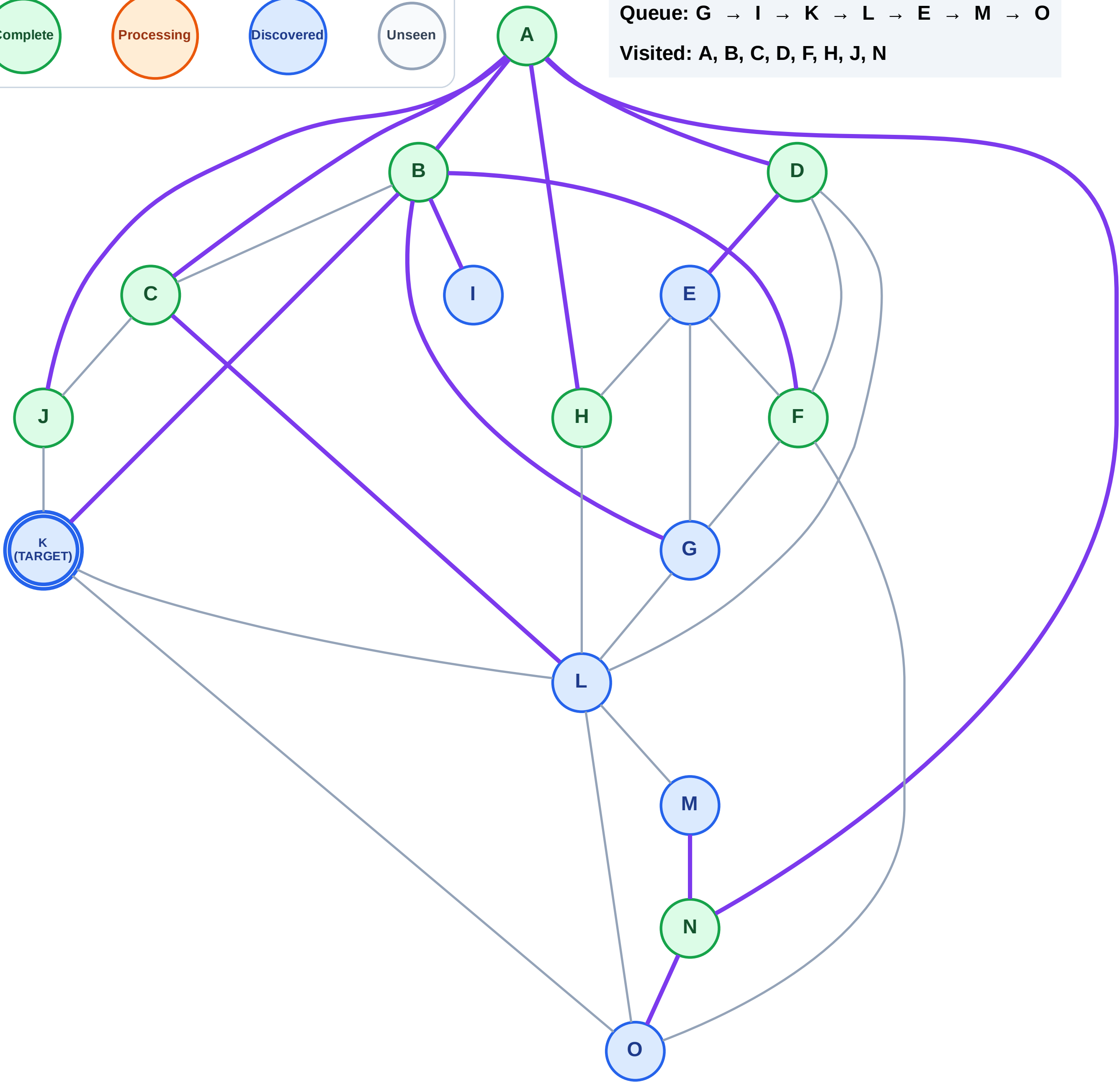
Processing

Discovered

Unseen

Queue: G → I → K → L → E → M → O

Visited: A, B, C, D, F, H, J, N



BFS Traversal

Step 18: Process node G

Legend

Complete

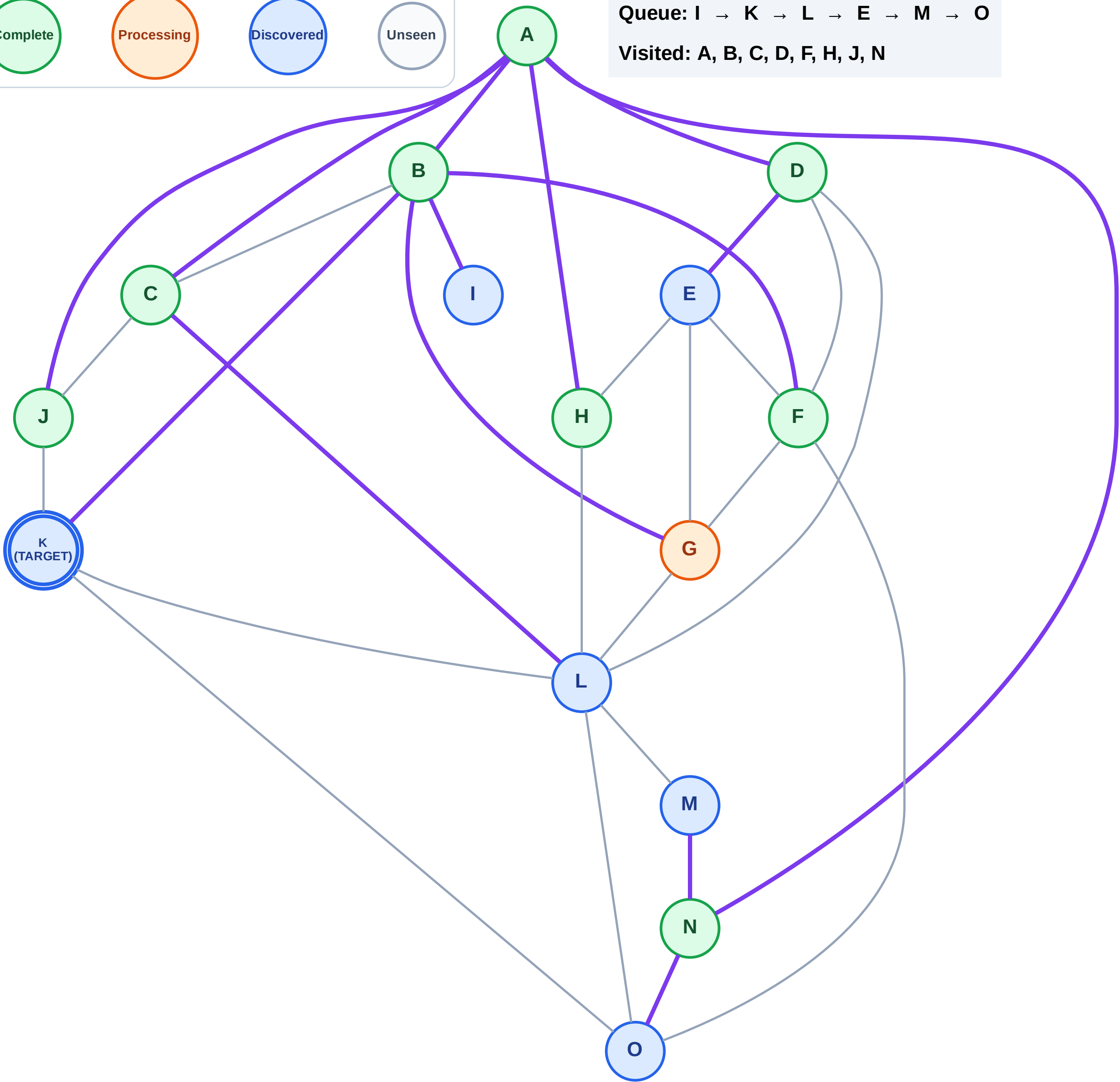
Processing

Discovered

Unseen

Queue: I → K → L → E → M → O

Visited: A, B, C, D, F, H, J, N



BFS Traversal

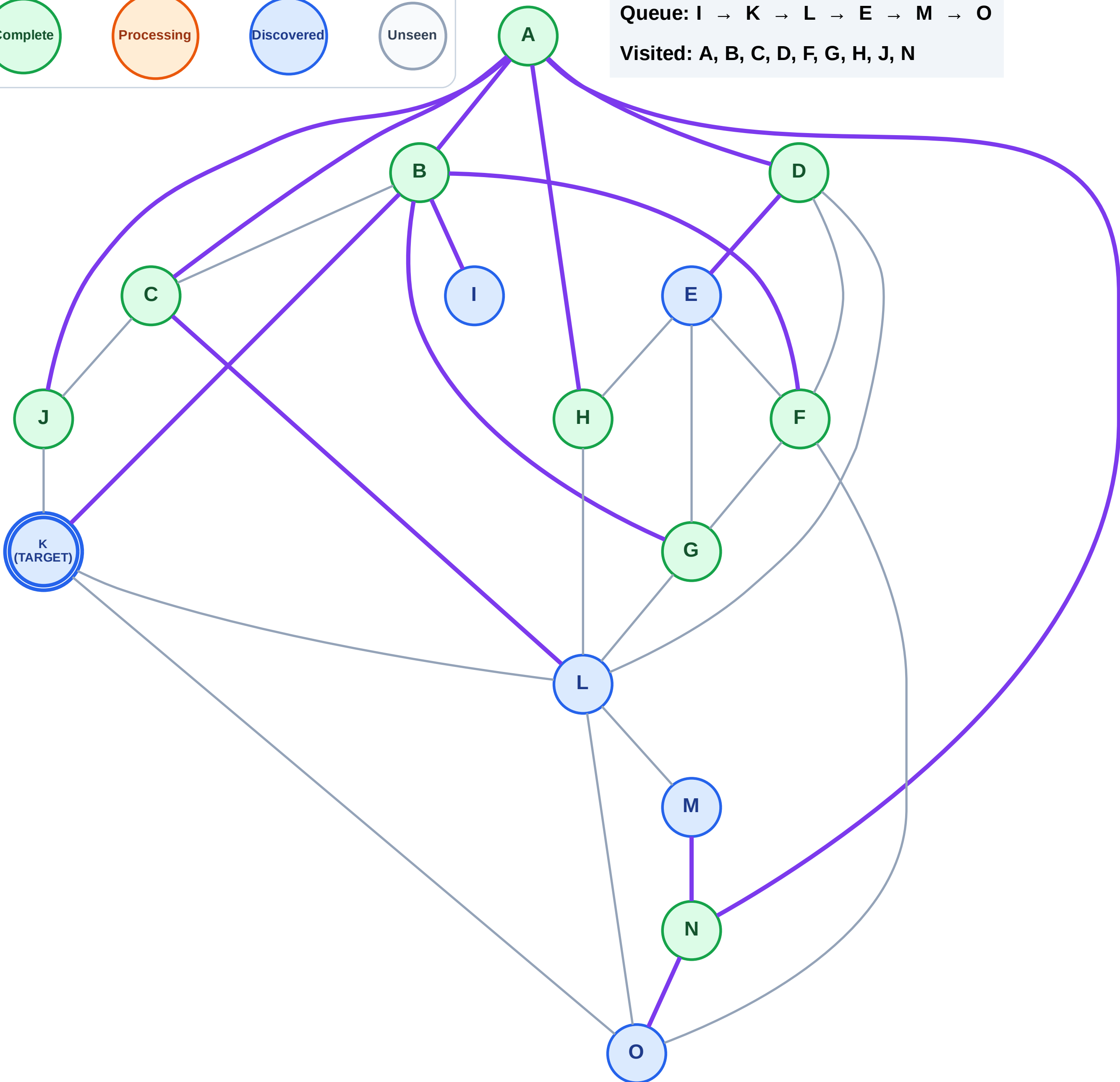
Step 19: No new neighbors from G

Legend



Queue: I → K → L → E → M → O

Visited: A, B, C, D, F, G, H, J, N



BFS Traversal

Step 20: Process node I

Legend

Complete

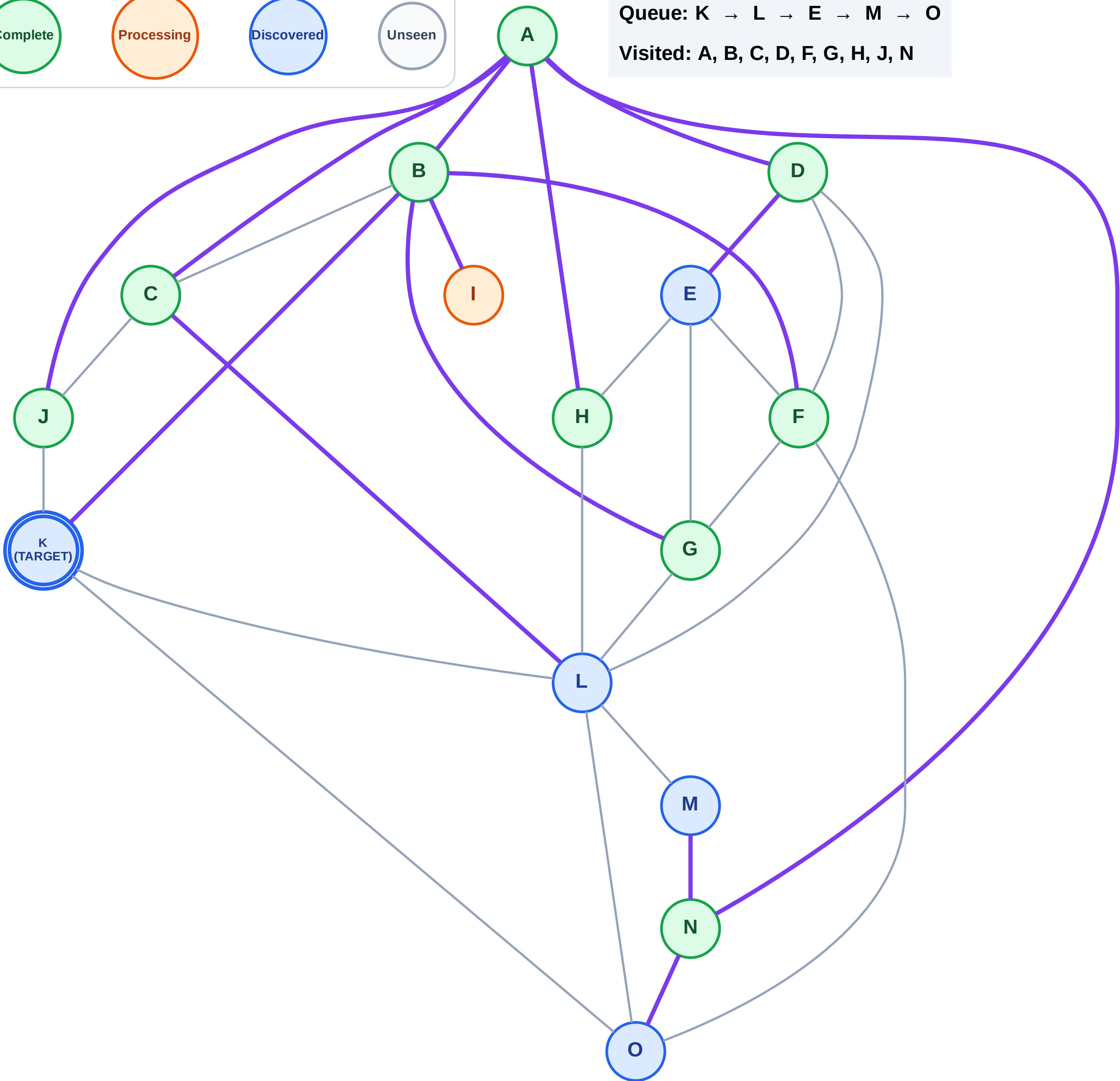
Processing

Discovered

Unseen

Queue: K → L → E → M → O

Visited: A, B, C, D, F, G, H, J, N



BFS Traversal

Step 21: No new neighbors from I

Legend

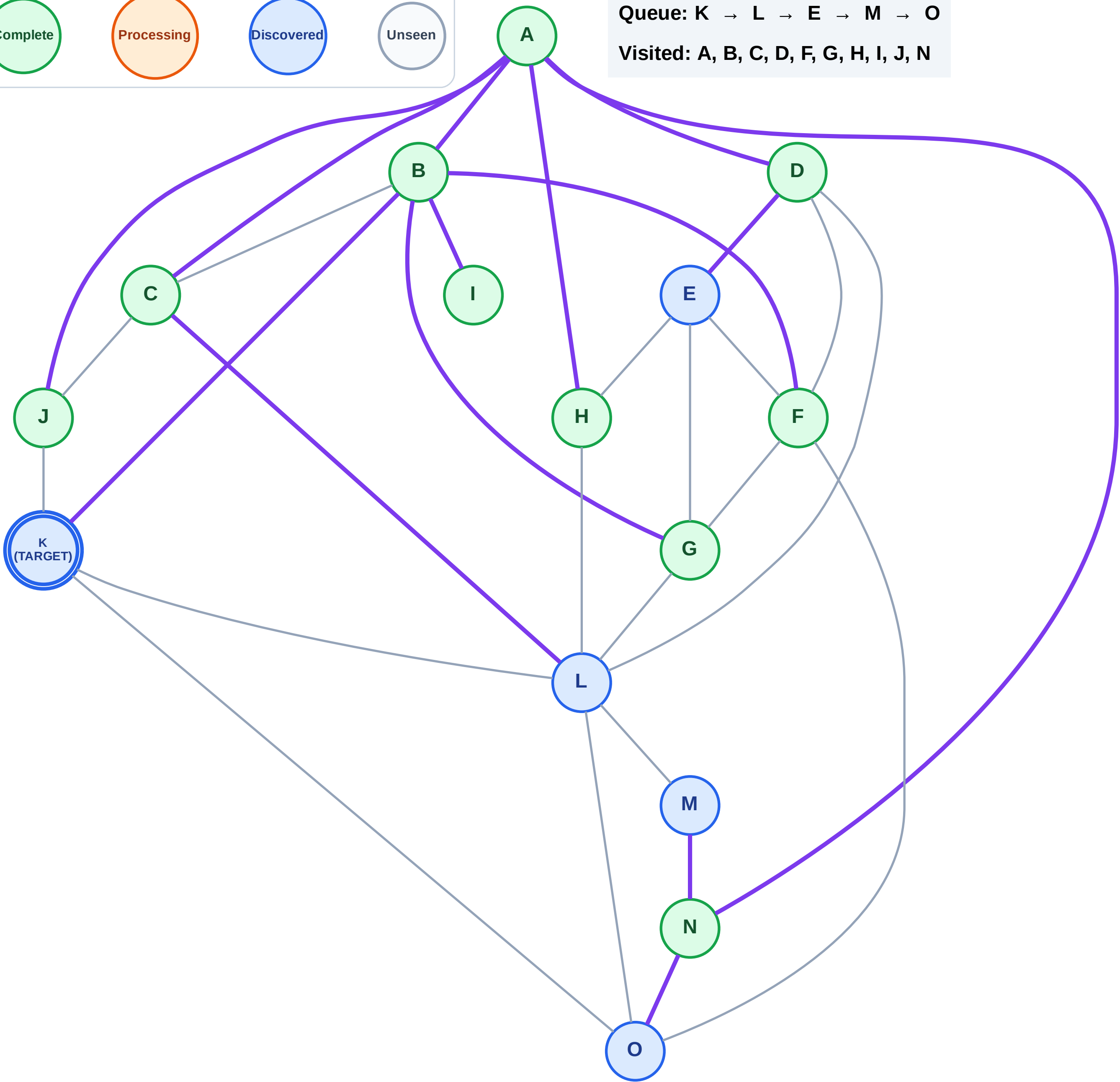
Complete

Processing

Discovered

Unseen

Queue: K → L → E → M → O
Visited: A, B, C, D, F, G, H, I, J, N



BFS Traversal

Step 22: Process node K

Legend

Complete

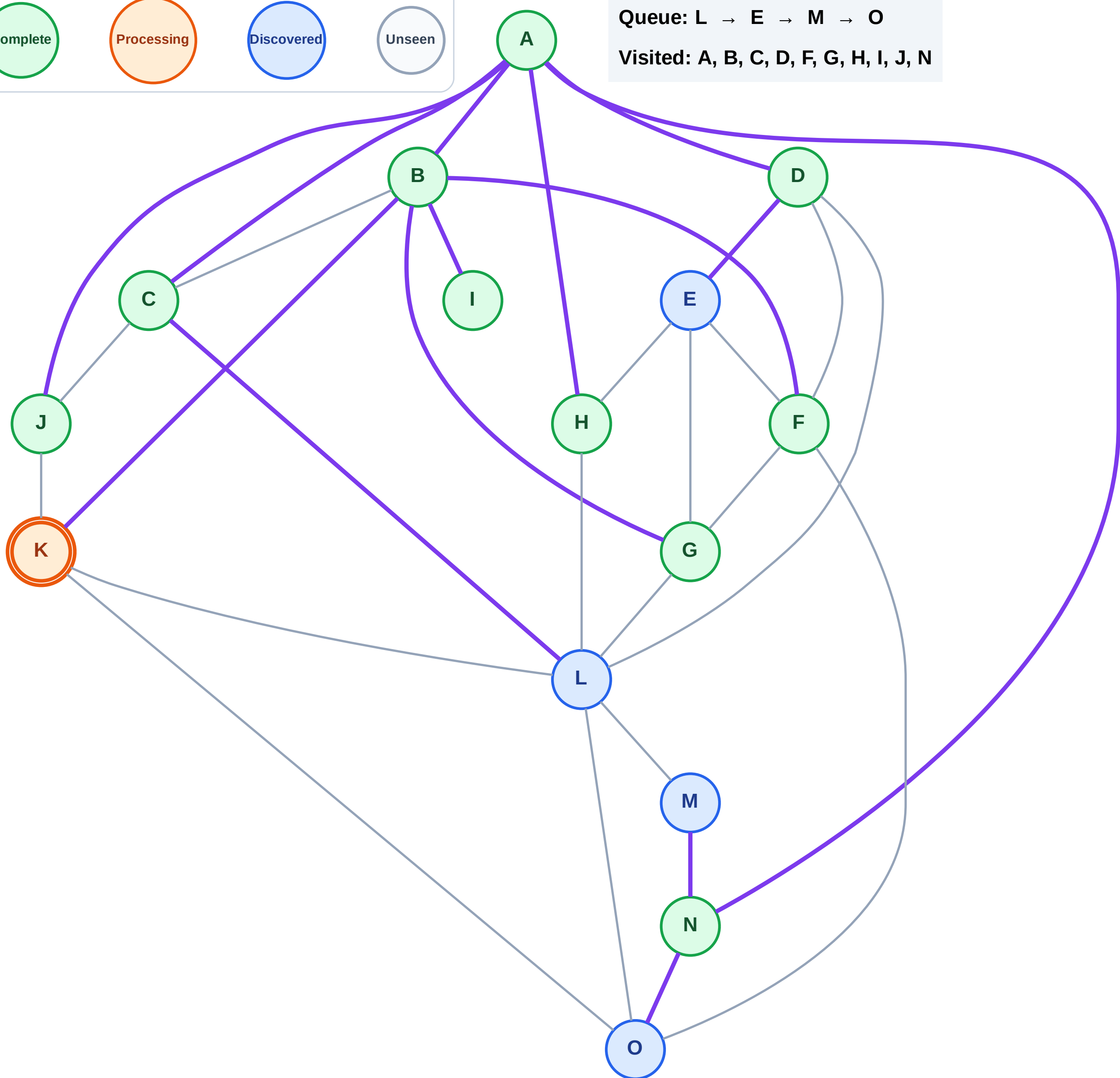
Processing

Discovered

Unseen

Queue: L → E → M → O

Visited: A, B, C, D, F, G, H, I, J, N



BFS Traversal

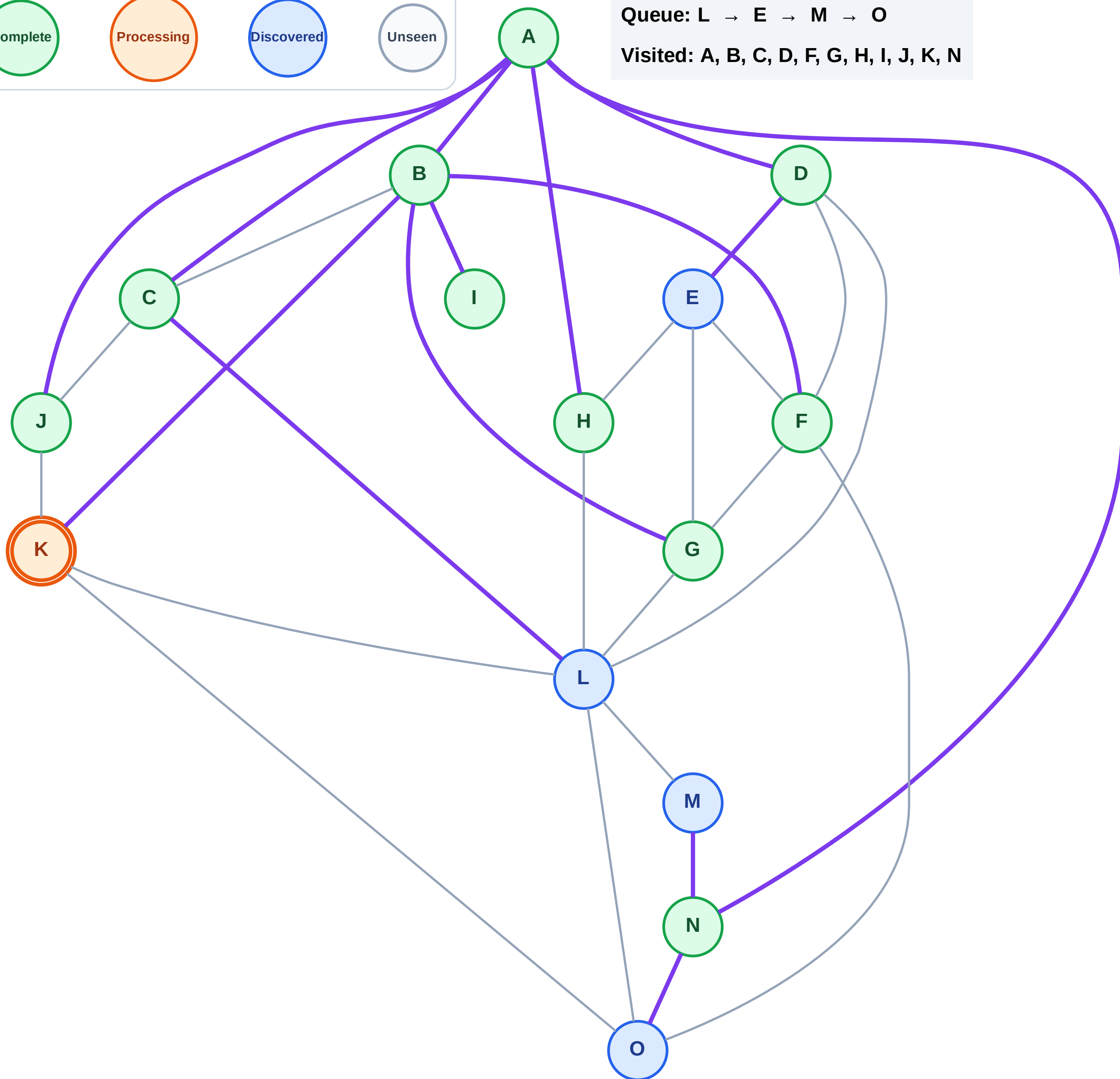
Step 23: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend



Queue: L → E → M → O

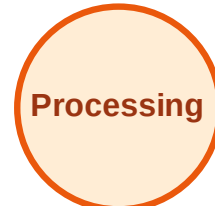
Visited: A, B, C, D, F, G, H, I, J, K, N



DFS Traversal

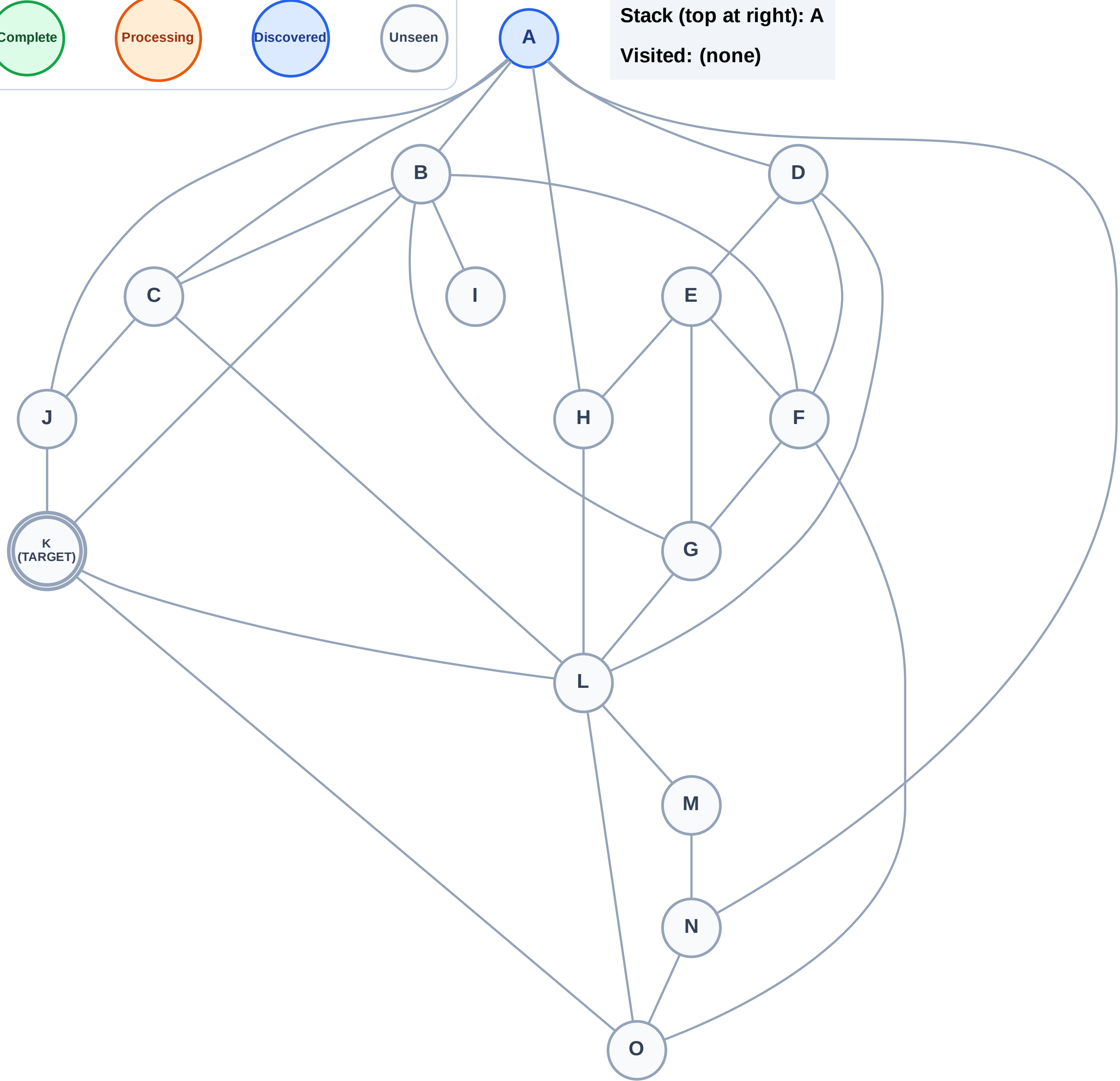
Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

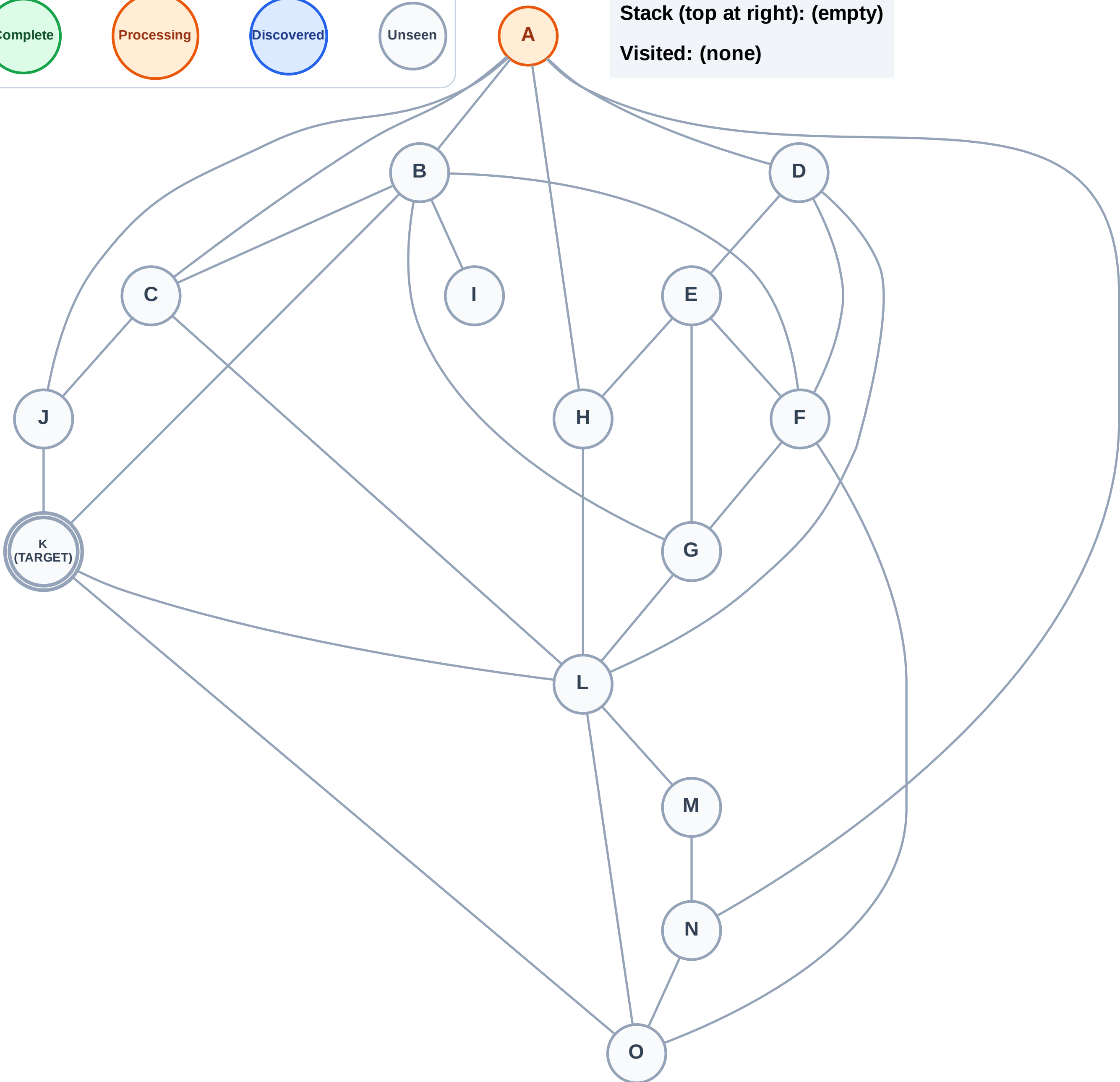
Step 2: Process node A

Legend



Stack (top at right): (empty)

Visited: (none)



DFS Traversal

Step 3: Discover N, J, H, D, C, B from A

Legend

Complete

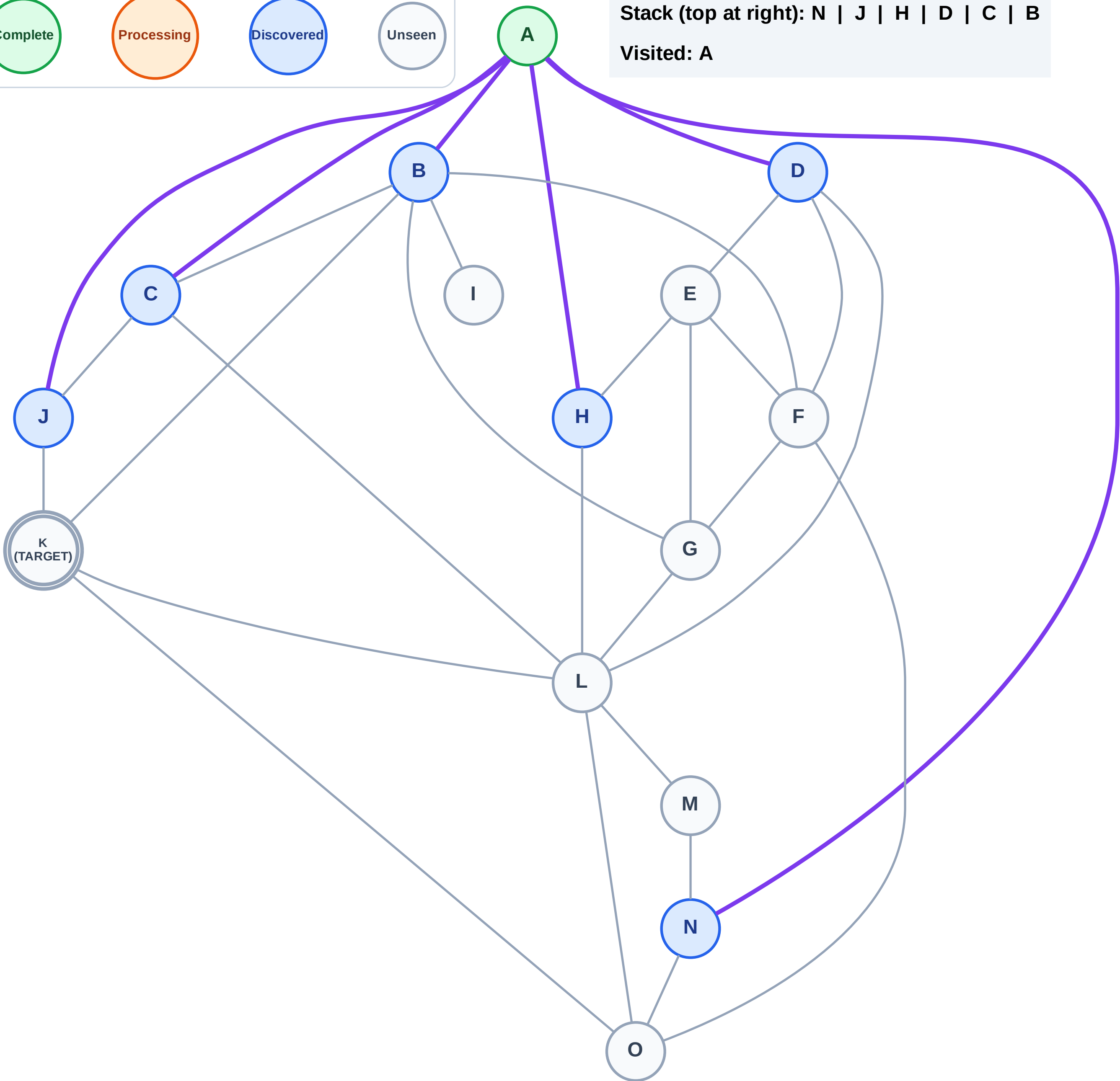
Processing

Discovered

Unseen

Stack (top at right): N | J | H | D | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

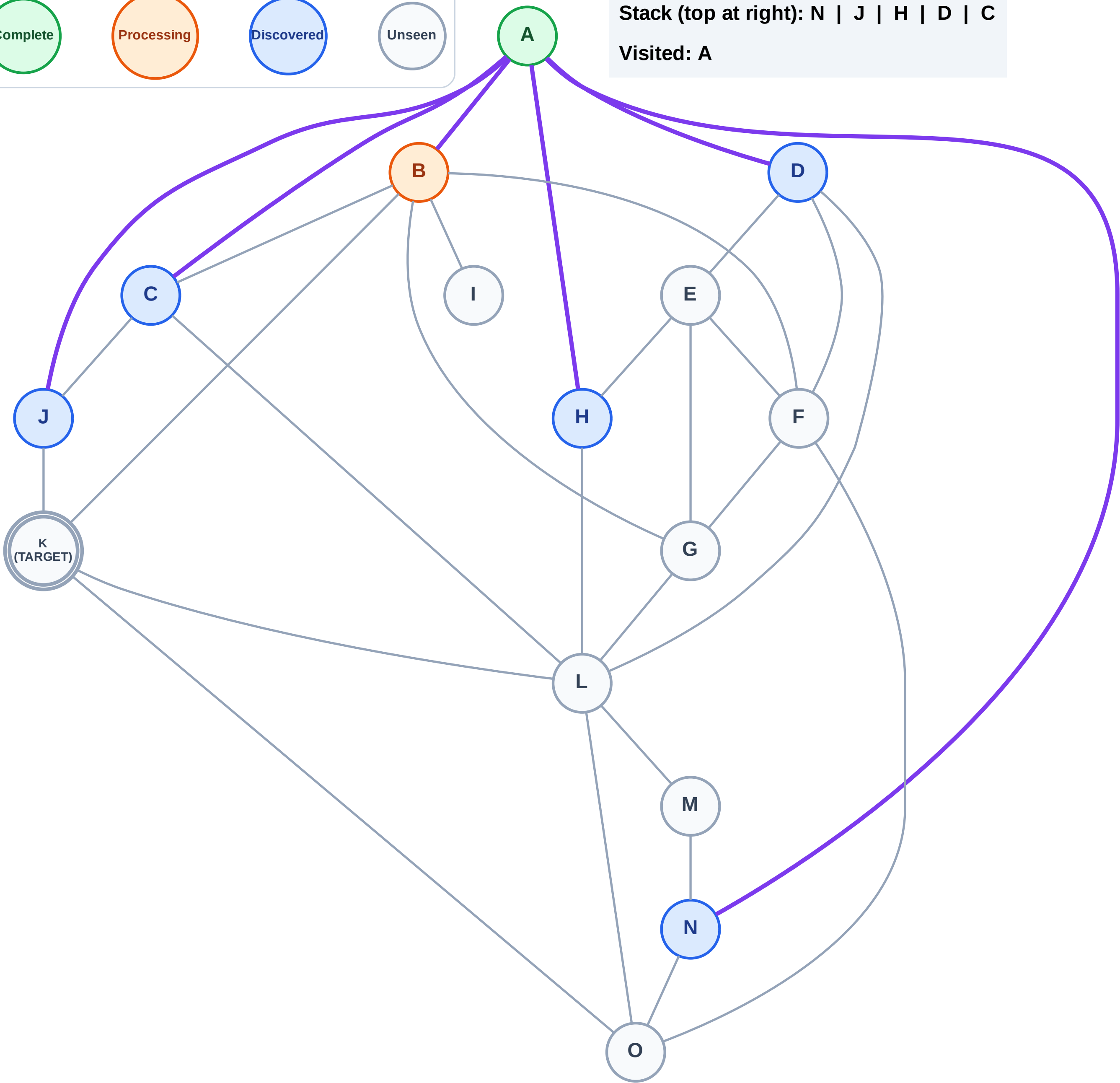
Processing

Discovered

Unseen

Stack (top at right): N | J | H | D | C

Visited: A



DFS Traversal

Step 5: Discover K, I, G, F from B

Legend

Complete

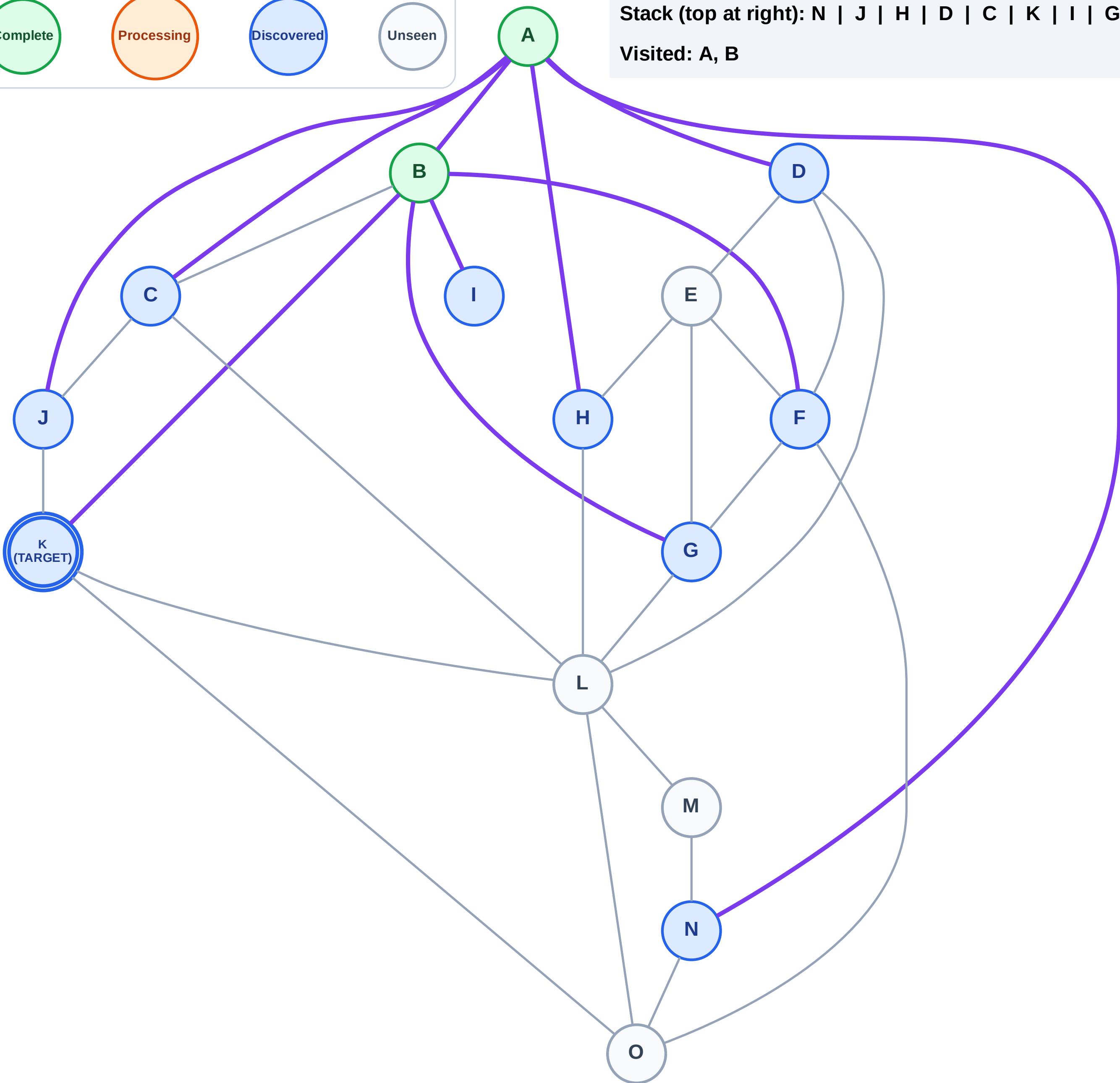
Processing

Discovered

Unseen

Stack (top at right): N | J | H | D | C | K | I | G | F

Visited: A, B



DFS Traversal

Step 6: Process node F

Legend

Complete

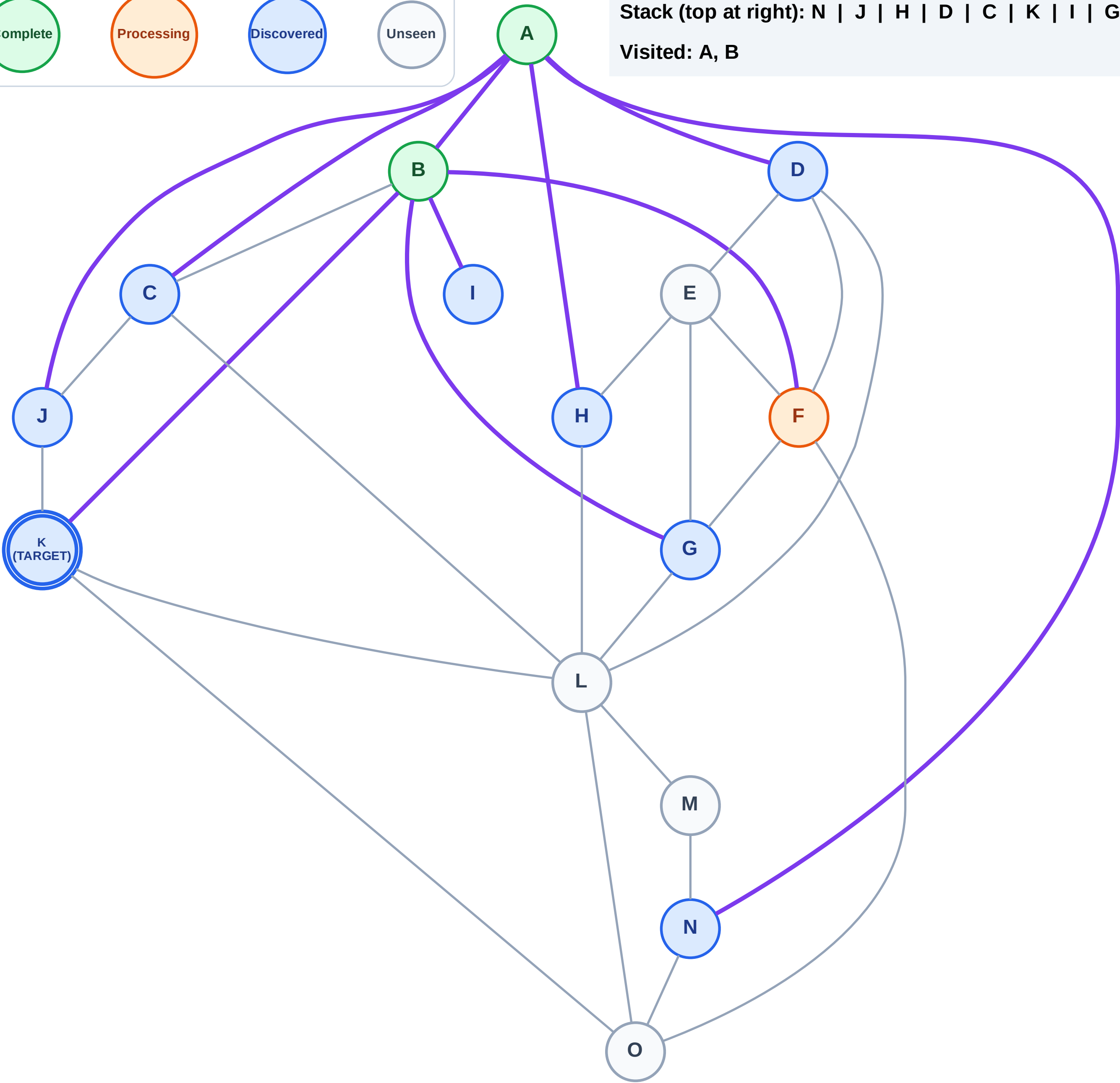
Processing

Discovered

Unseen

Stack (top at right): N | J | H | D | C | K | I | G

Visited: A, B



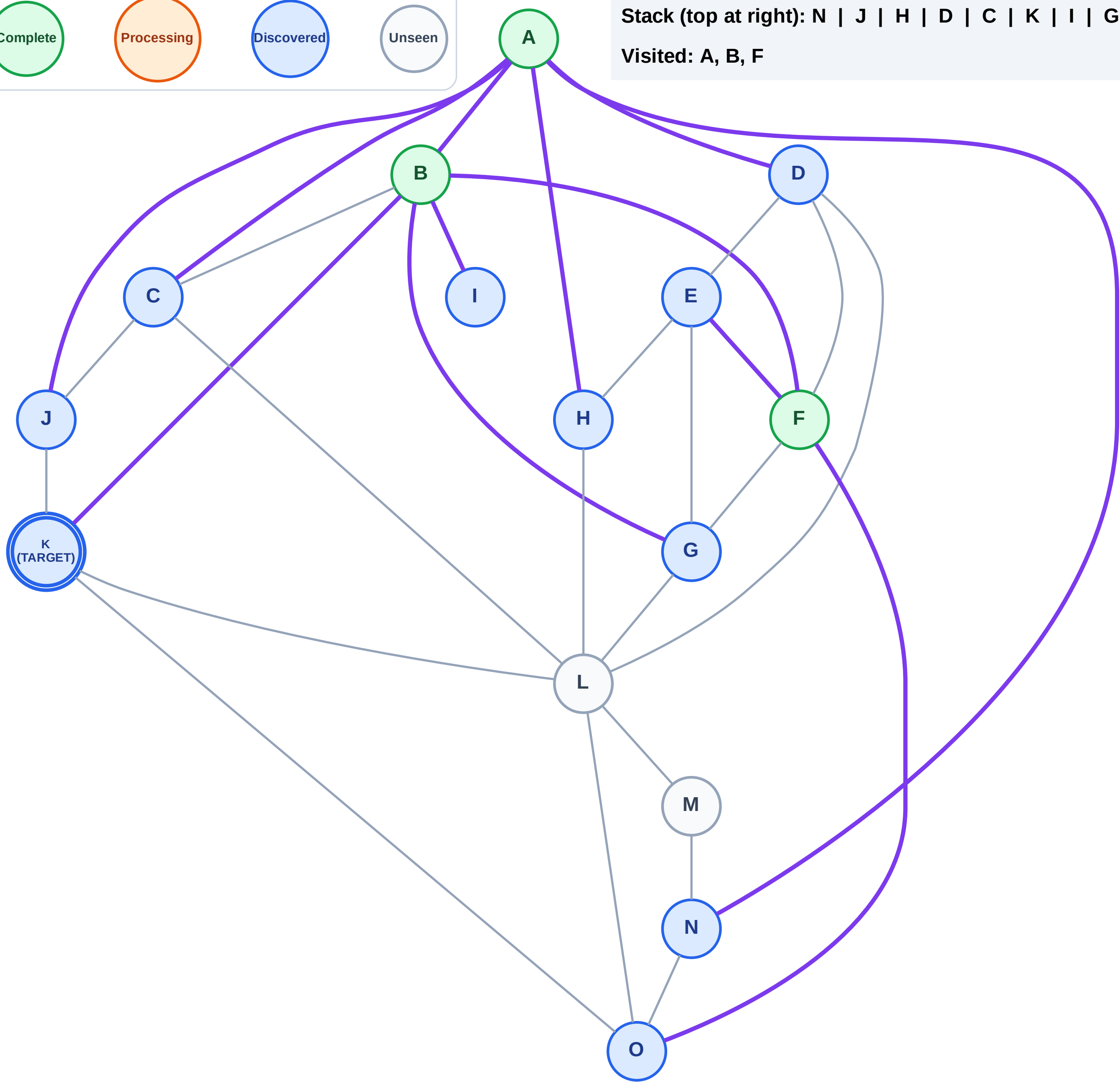
Step 7: Discover O, E from F

Stack (top at right): N | J | H | D | C | K | I | G | O | E

Legend

Processing

Unseen



DFS Traversal

Step 8: Process node E

Legend

Complete

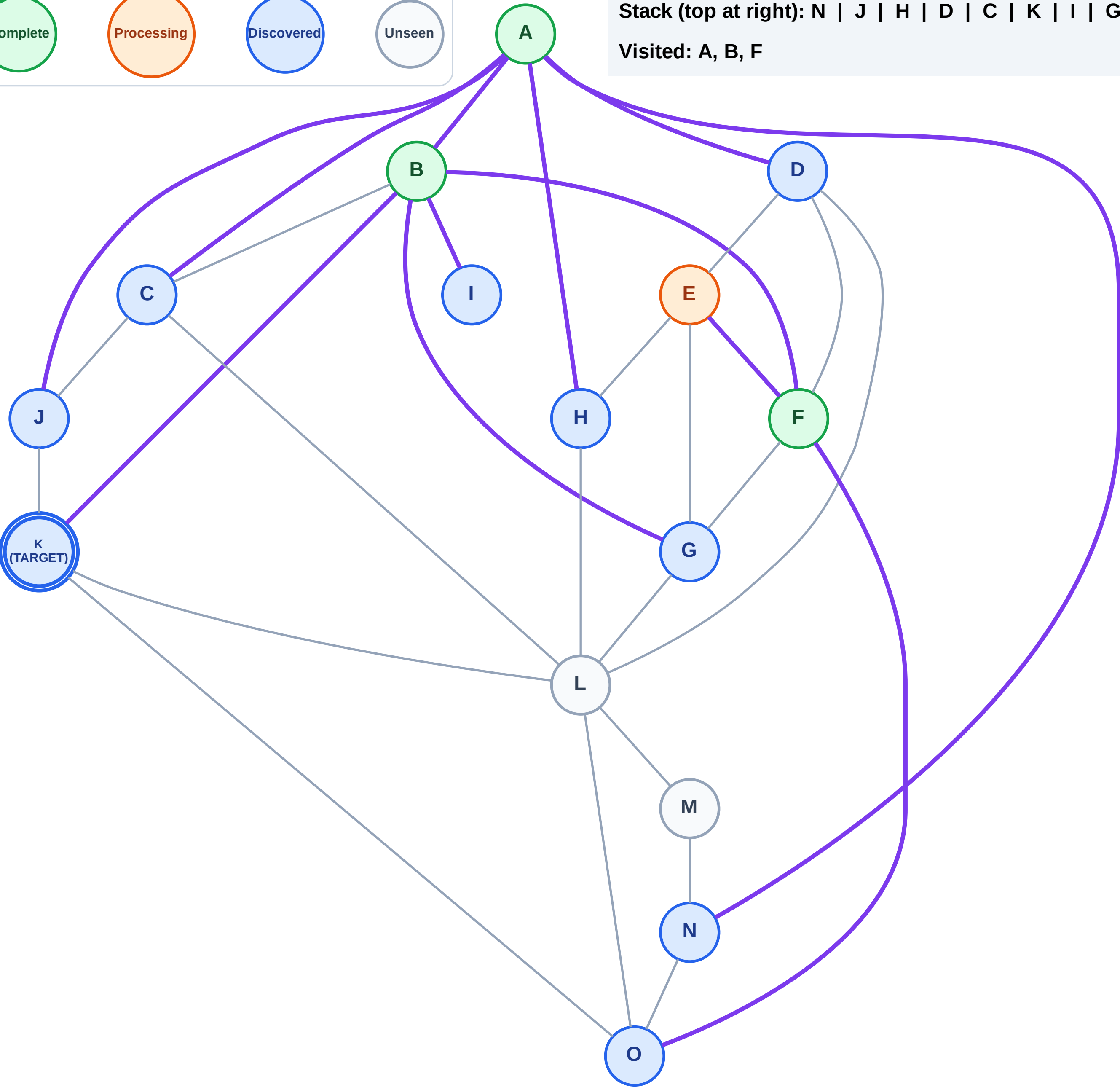
Processing

Discovered

Unseen

Stack (top at right): N | J | H | D | C | K | I | G | O

Visited: A, B, F



DFS Traversal

Step 9: No new neighbors from E

Legend

Complete

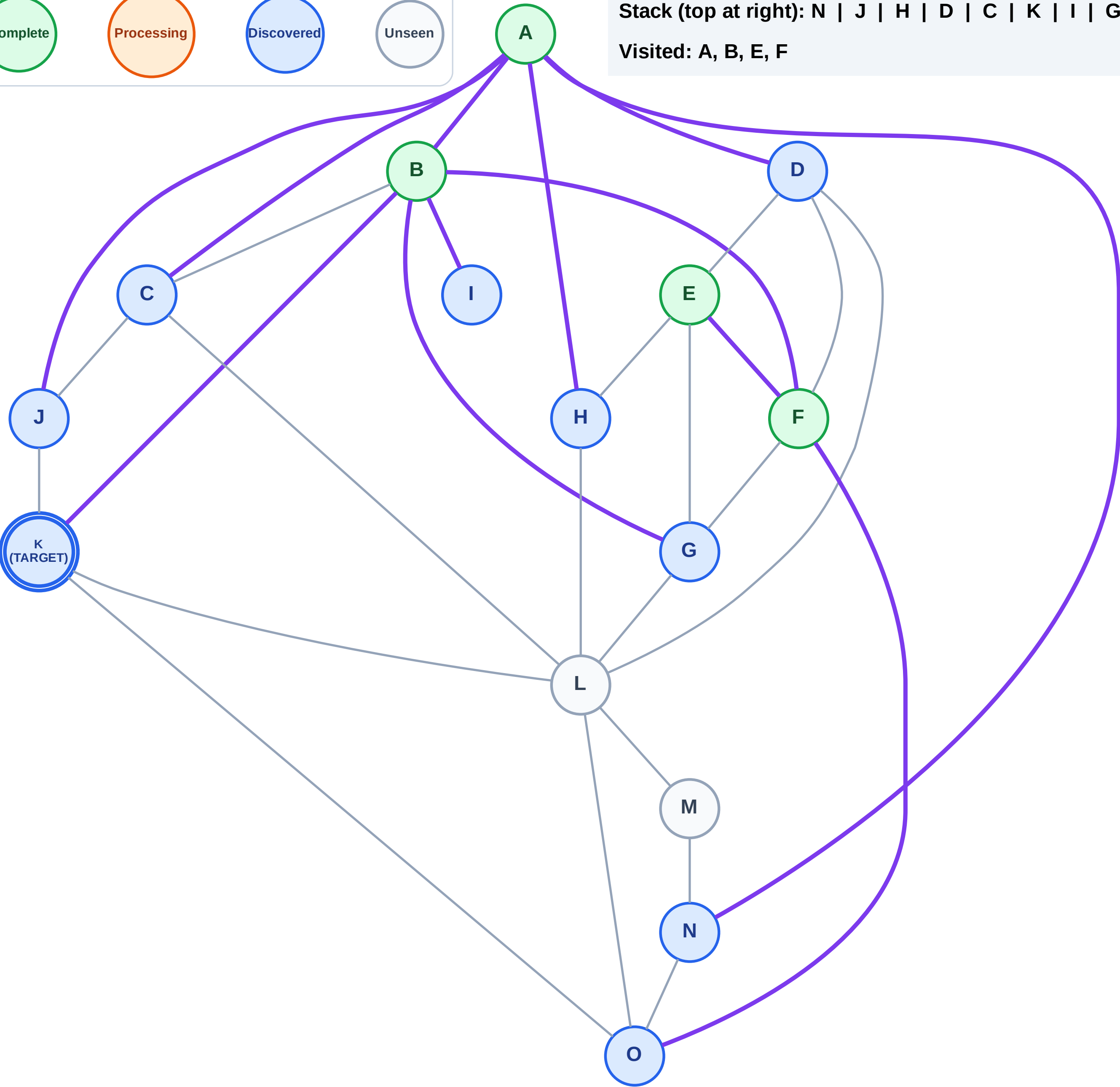
Processing

Discovered

Unseen

Stack (top at right): N | J | H | D | C | K | I | G | O

Visited: A, B, E, F



DFS Traversal

Step 10: Process node O

Legend

Complete

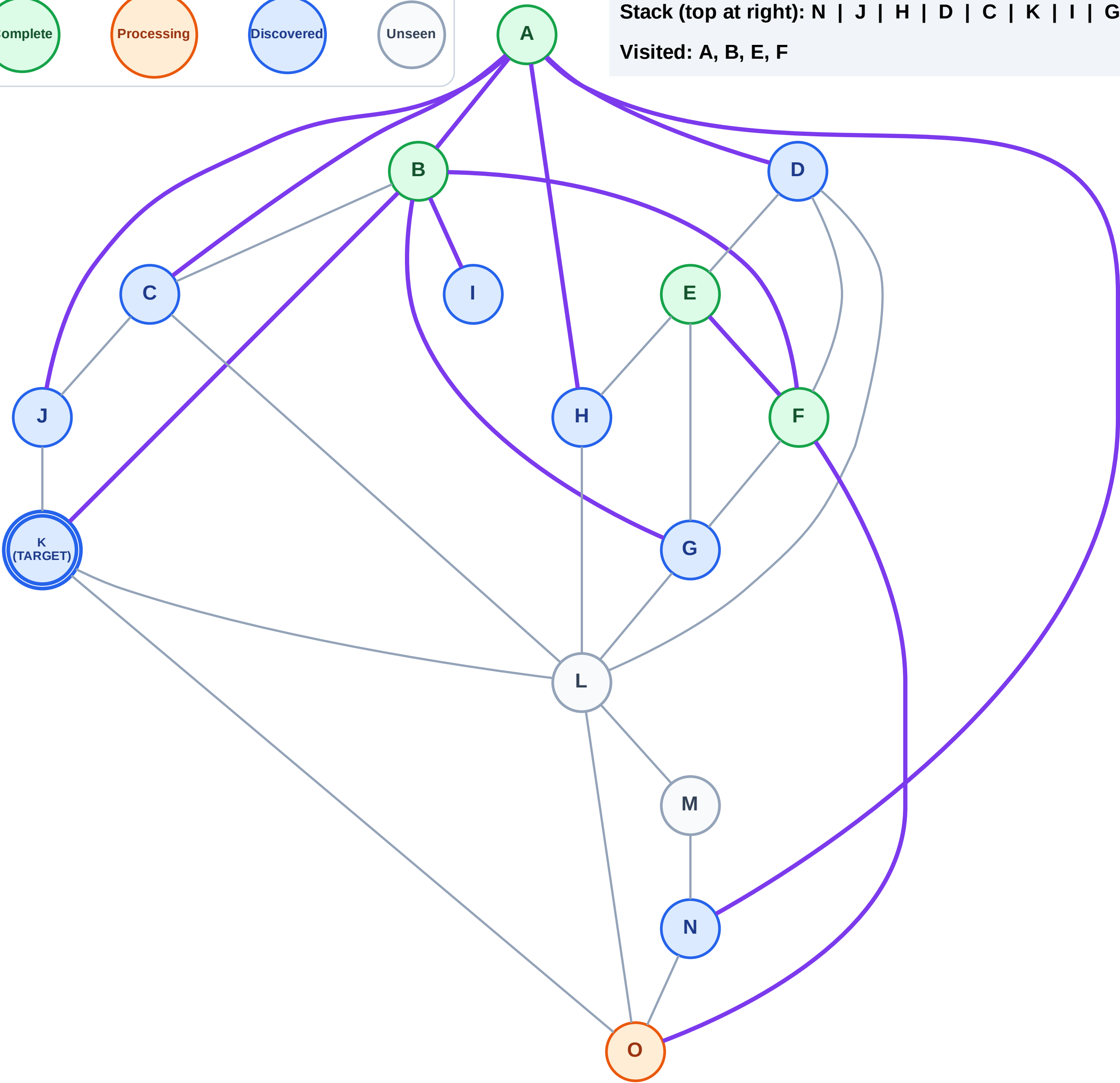
Processing

Discovered

Unseen

Stack (top at right): N | J | H | D | C | K | I | G

Visited: A, B, E, F



DFS Traversal

Step 11: Discover L from O

Legend

Complete

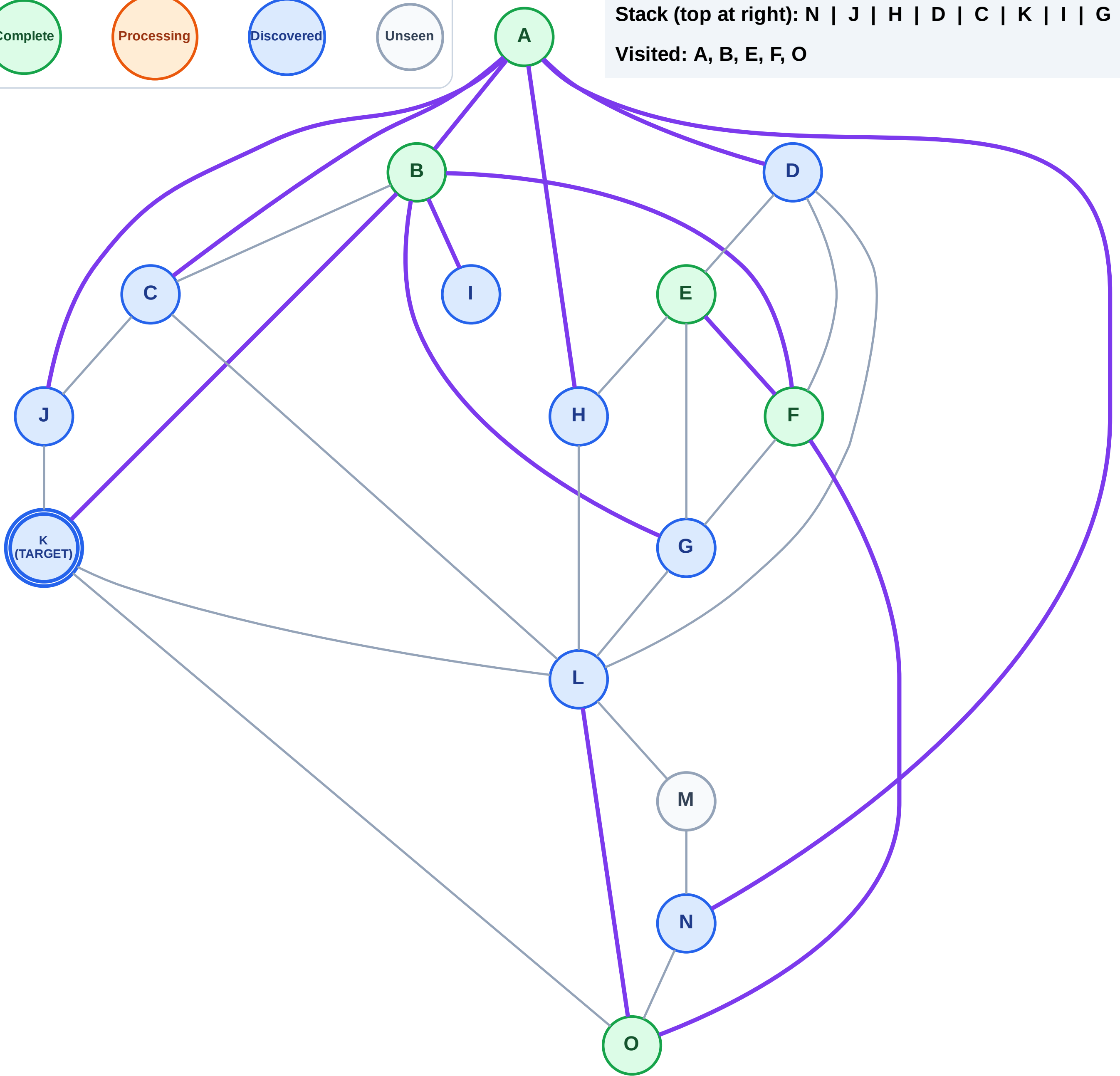
Processing

Discovered

Unseen

Stack (top at right): N | J | H | D | C | K | I | G | L

Visited: A, B, E, F, O



DFS Traversal

Step 12: Process node L

Legend

Complete

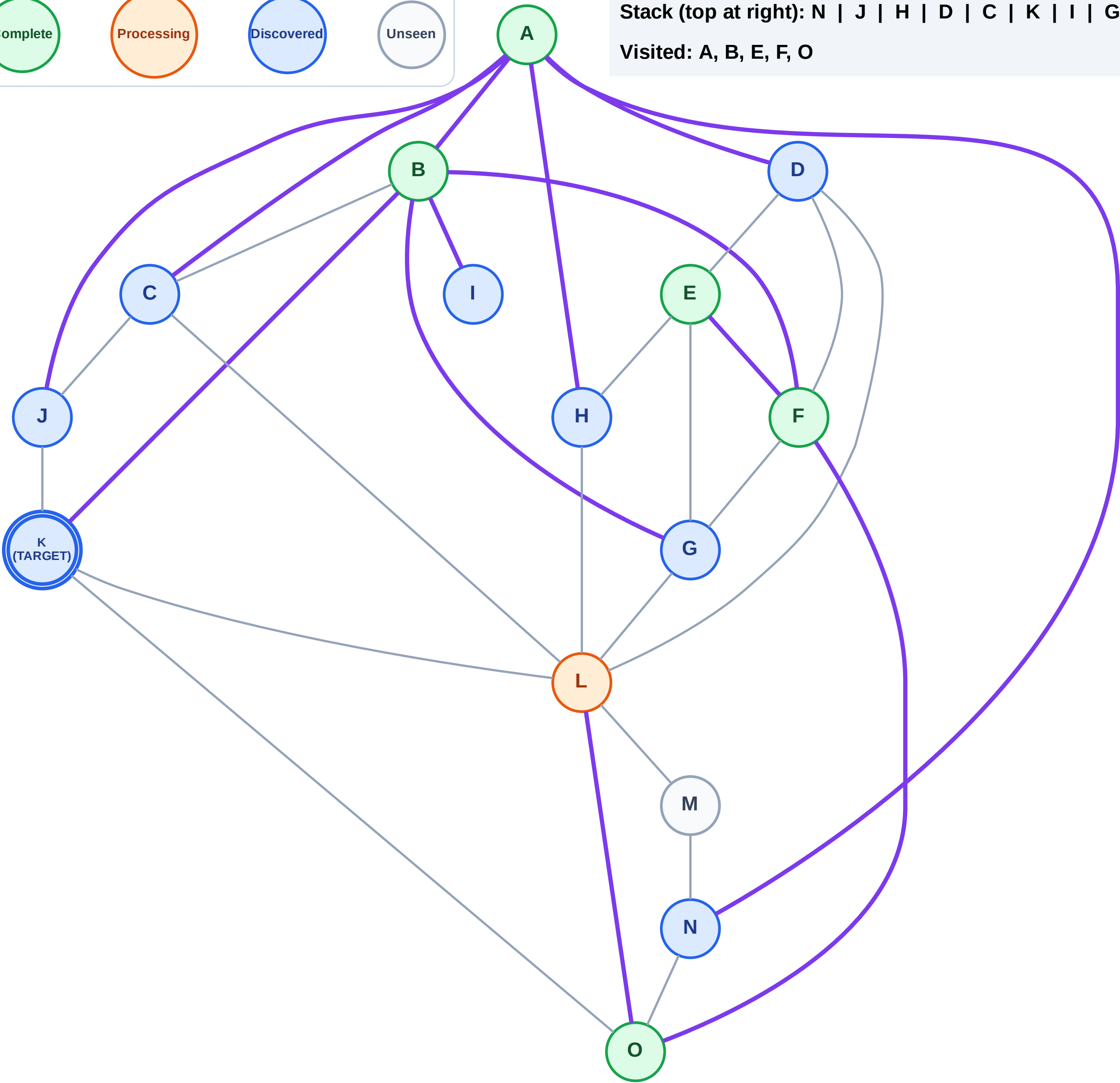
Processing

Discovered

Unseen

Stack (top at right): N | J | H | D | C | K | I | G

Visited: A, B, E, F, O



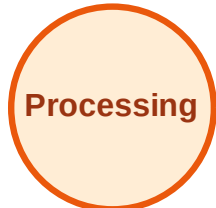
DFS Traversal

Step 13: Discover M from L

Legend



Complete



Processing



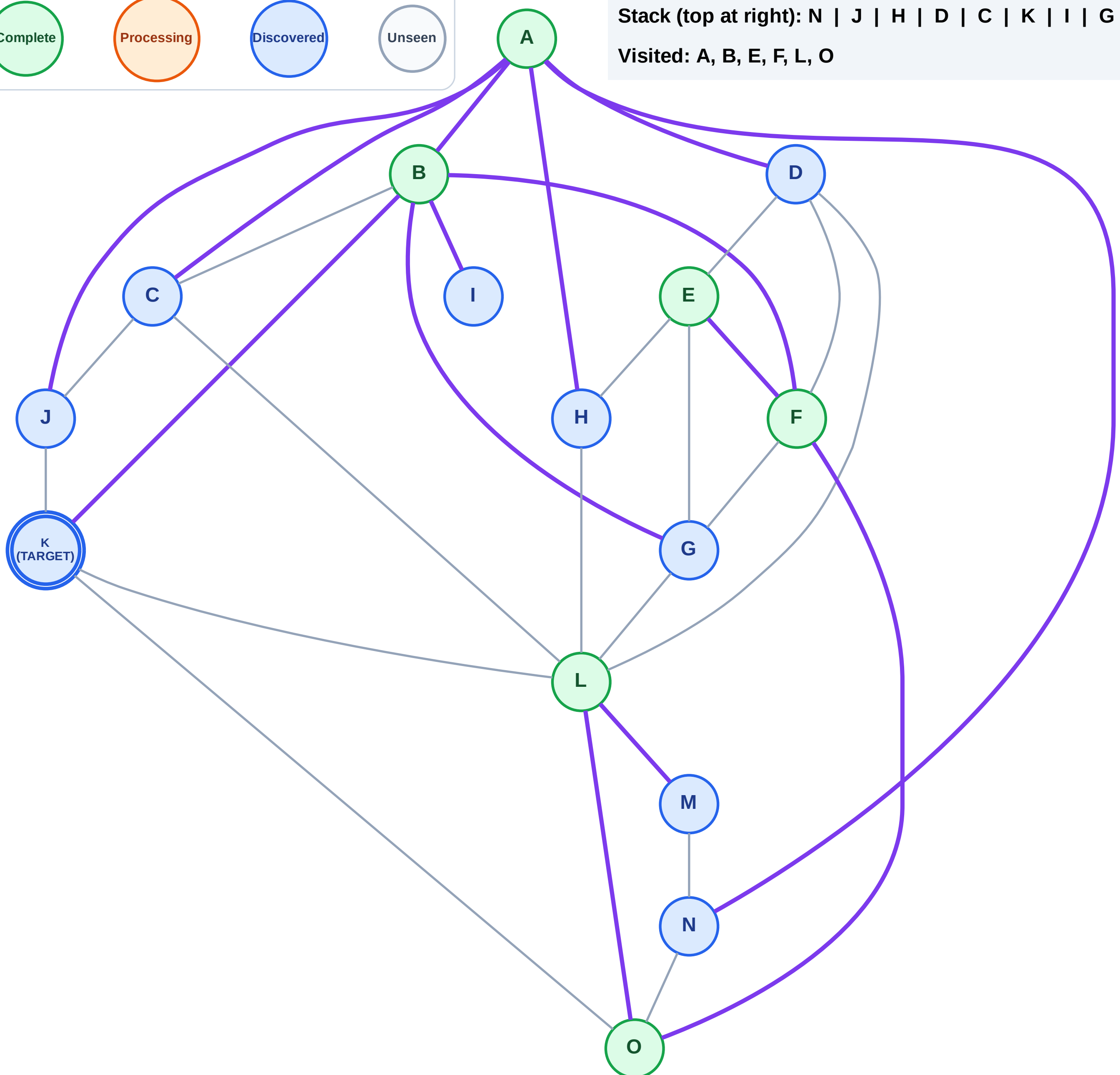
Discovered



Unseen

Stack (top at right): N | J | H | D | C | K | I | G | M

Visited: A, B, E, F, L, O



DFS Traversal

Step 14: Process node M

Legend

Complete

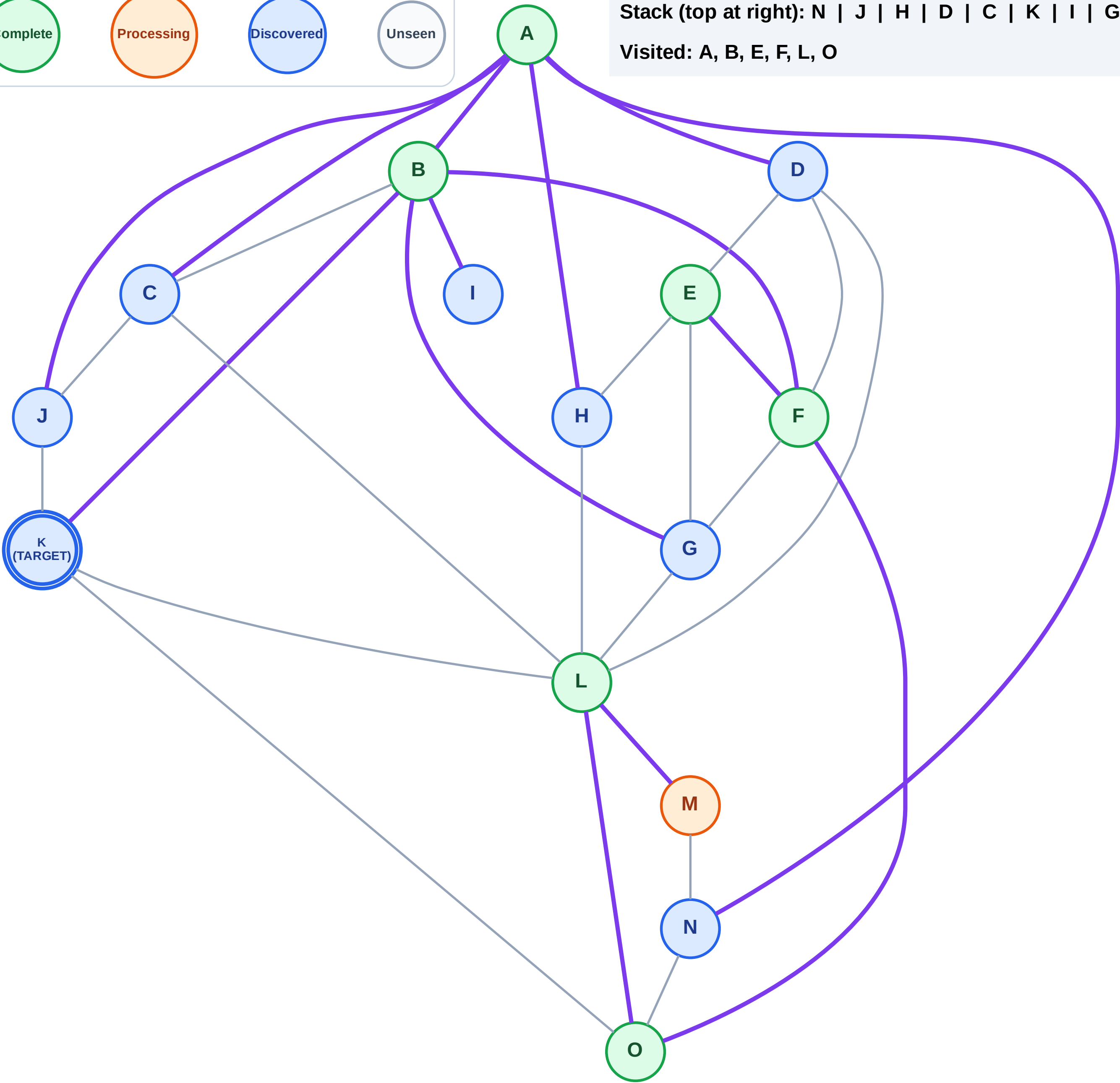
Processing

Discovered

Unseen

Stack (top at right): N | J | H | D | C | K | I | G

Visited: A, B, E, F, L, O



DFS Traversal

Step 15: No new neighbors from M

Legend

Complete

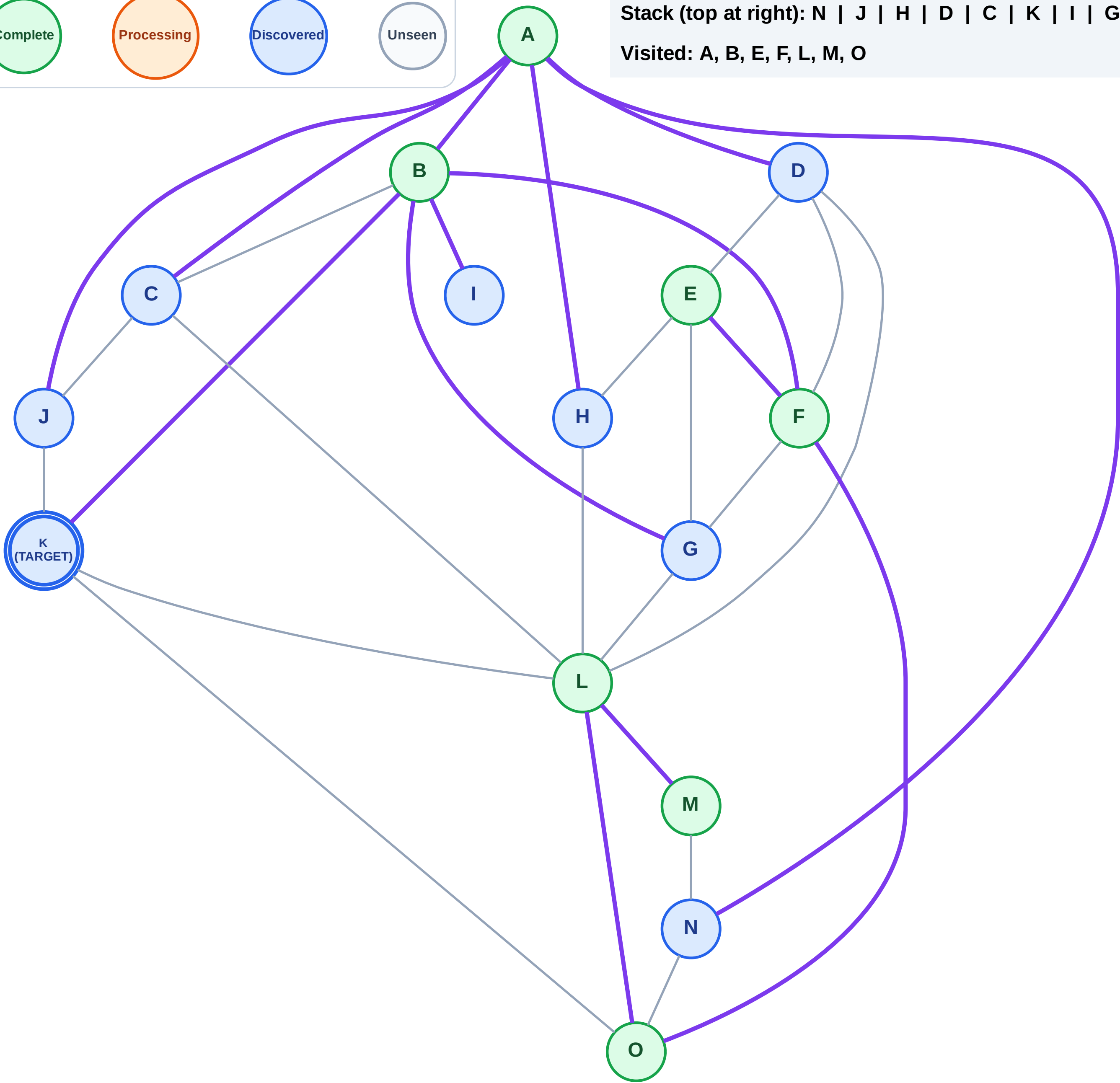
Processing

Discovered

Unseen

Stack (top at right): N | J | H | D | C | K | I | G

Visited: A, B, E, F, L, M, O



DFS Traversal

Step 16: Process node G

Legend

Complete

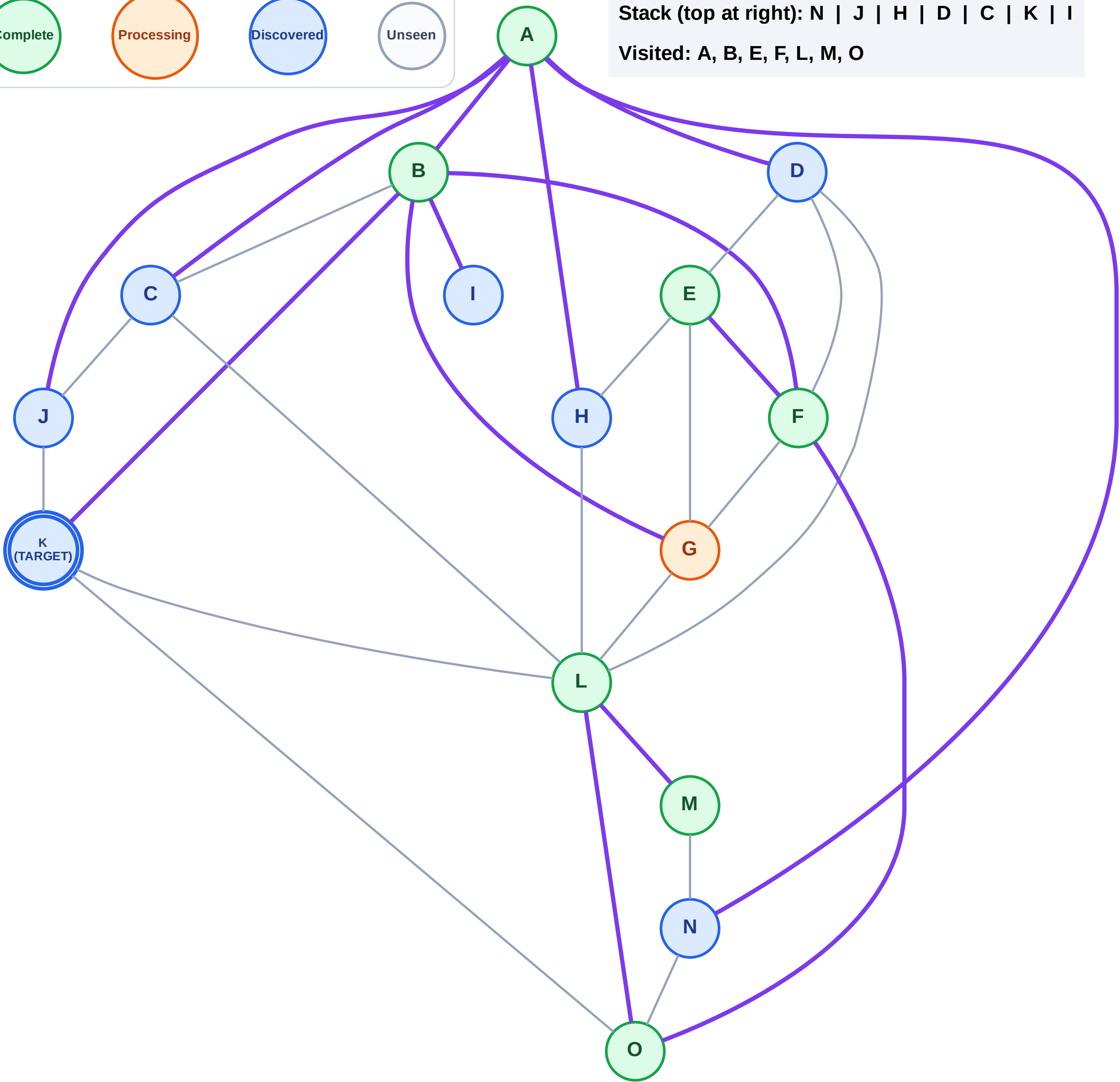
Processing

Discovered

Unseen

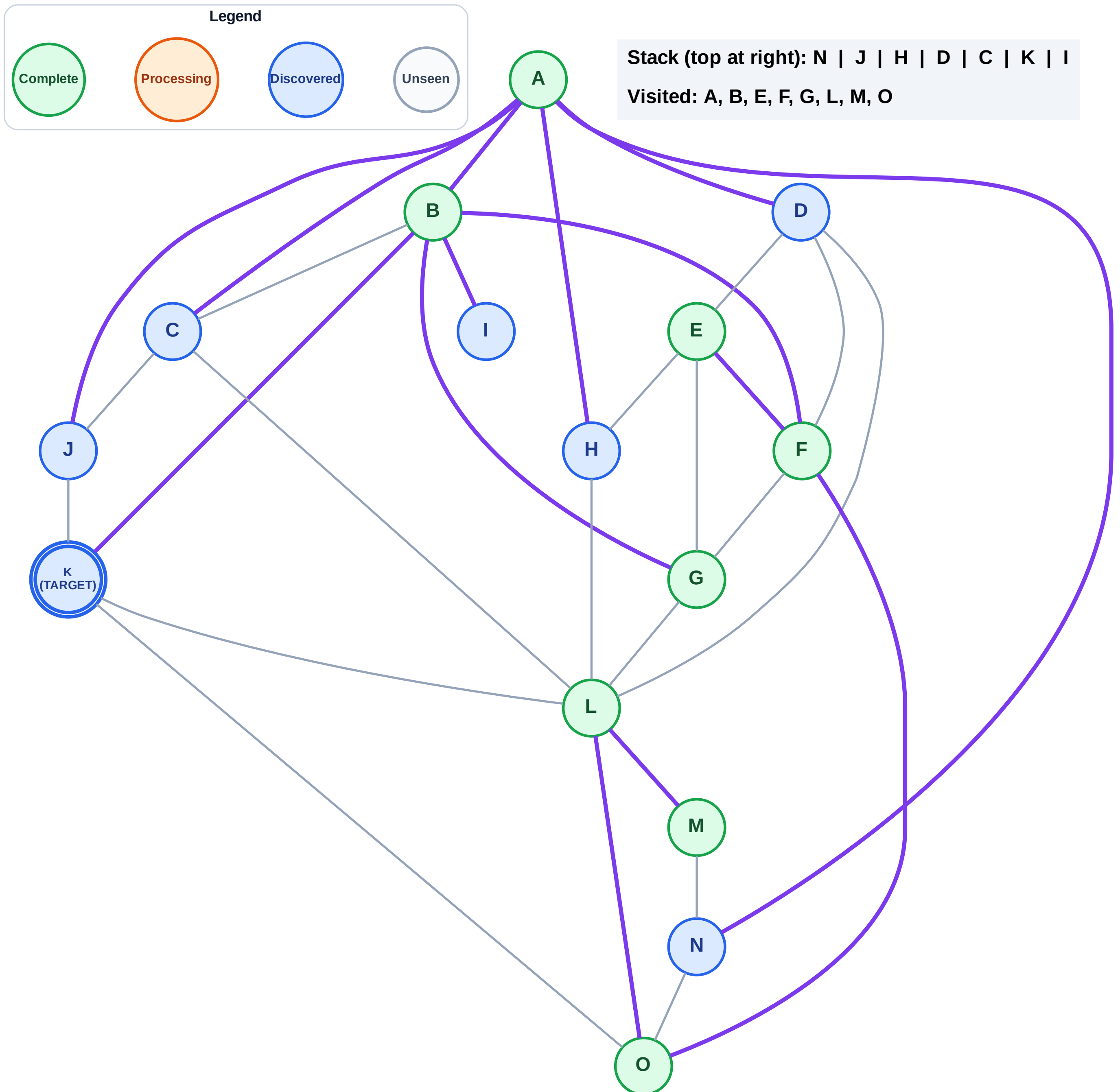
Stack (top at right): N | J | H | D | C | K | I

Visited: A, B, E, F, L, M, O



Step 17: No new neighbors from G

Step 17: No new neighbors from G



DFS Traversal

Step 18: Process node I

Legend

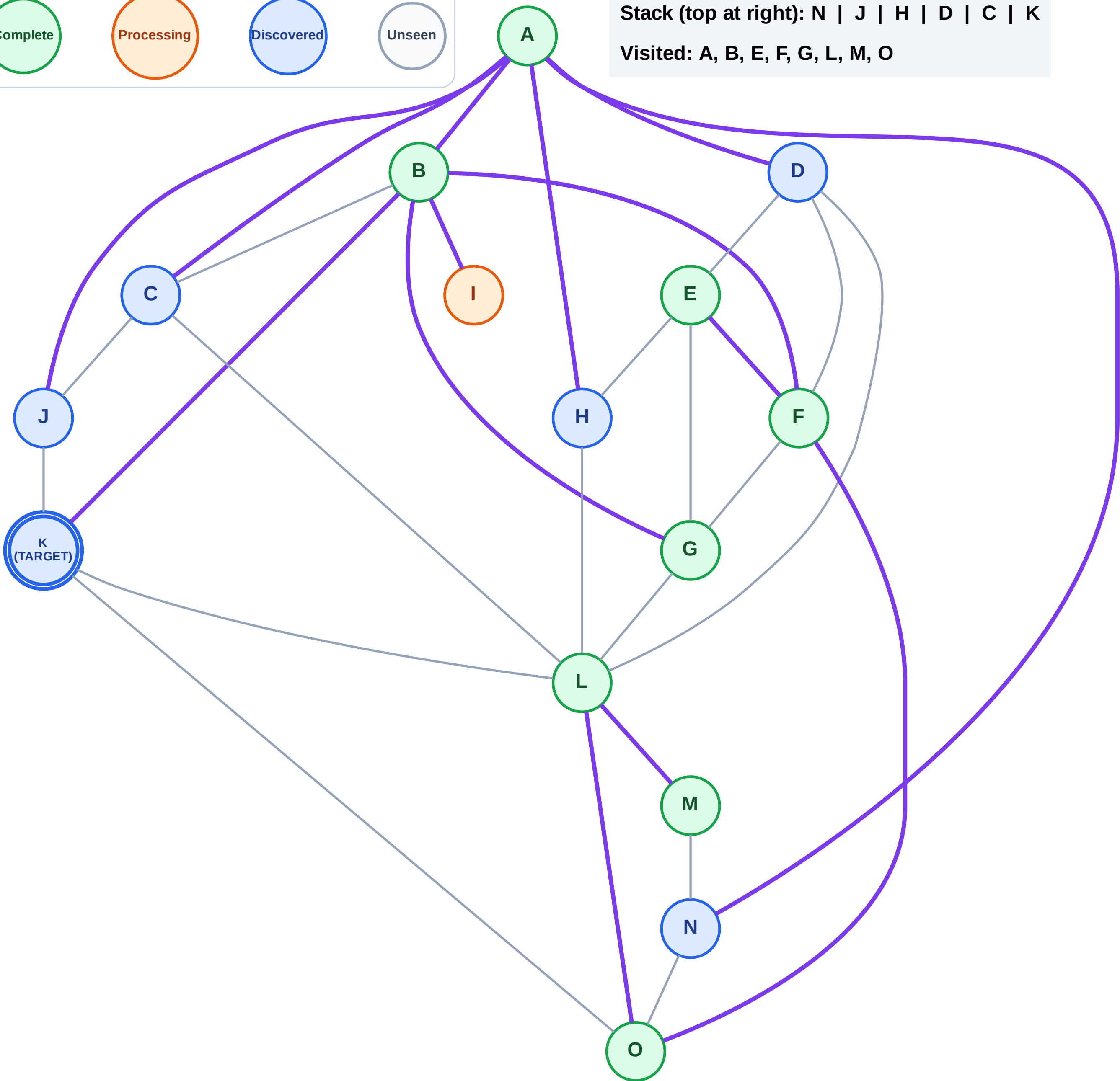
Complete

Processing

Discovered

Unseen

Stack (top at right): N | J | H | D | C | K
Visited: A, B, E, F, G, L, M, O



DFS Traversal

Step 19: No new neighbors from I

Legend

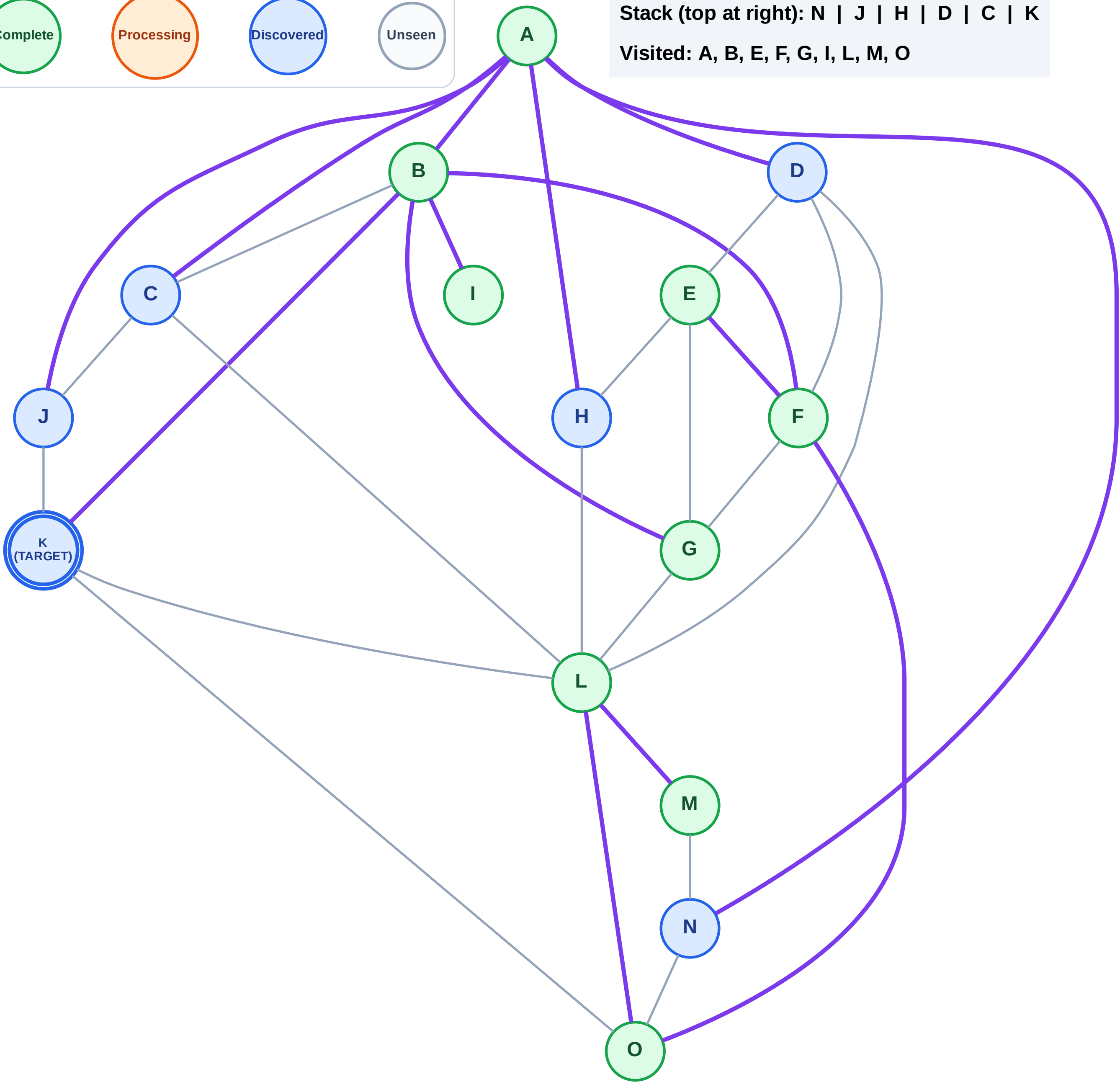
Complete

Processing

Discovered

Unseen

Stack (top at right): N | J | H | D | C | K
Visited: A, B, E, F, G, I, L, M, O



DFS Traversal

Step 20: Process node K

Legend

Complete

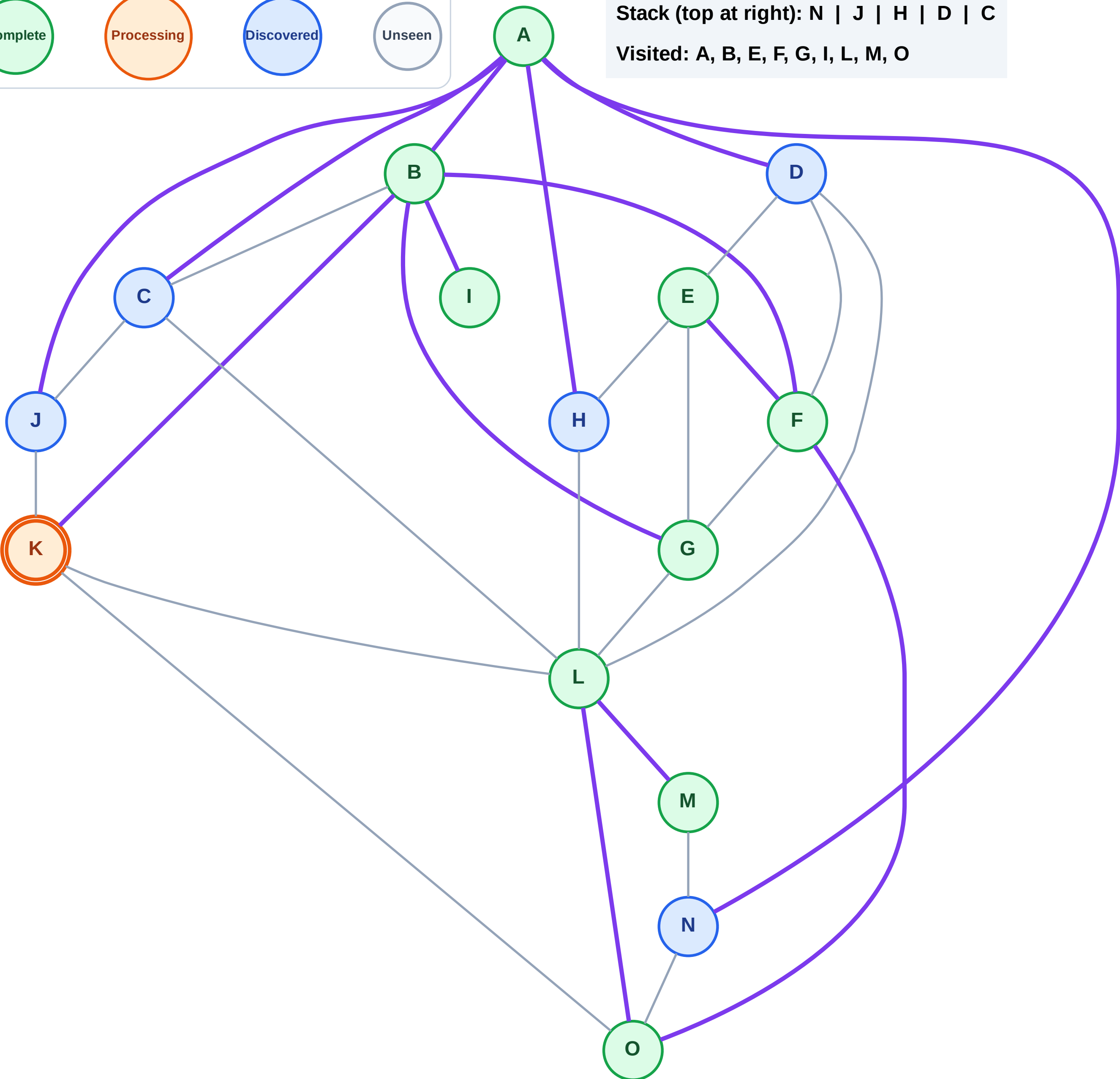
Processing

Discovered

Unseen

Stack (top at right): N | J | H | D | C

Visited: A, B, E, F, G, I, L, M, O



DFS Traversal

Step 21: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend



Complete



Processing



Discovered



Unseen

Stack (top at right): N | J | H | D | C

Visited: A, B, E, F, G, I, K, L, M, O

